Benchmark run full detail

Run ID: 20260429T001722Z_pure_full

Generation model: gemini / gemini-2.5-pro

Standard validator: gemini / gemini-2.5-flash

Layer 1 uses sentence-transformer greedy one-to-one matching at threshold 0.55. Layer 2 uses a checklist-based Gemini judge with strict and weighted scoring.

Document: 0000_cctns

Part A - Document header

Document ID	0000_cctns
Generation model	gemini / gemini-2.5-pro
Standard validator	gemini / gemini-2.5-flash
Dialogue method	N/A
Question algorithm	N/A
Source requirement count	100
Generated requirement count	91
Matched count	72
Clarification rounds	requested 2 / used 2
Clarification chunks	2
Semantic P / R / F1	0.7912 / 0.7200 / 0.7539
LLM strict P / R / F1	0.4835 / 0.4400 / 0.4607
LLM weighted P / R / F1	0.6264 / 0.5700 / 0.5969
Evidence / Propositions / Gap	105 / 91 / 0
Grounded after rewrite	91

Part B - Generated dialogue transcript

[SYSTEM] To start, what is the overall purpose and scope of this system?

[USER] This project is based on the 0000 - cctns system, and the goal is to capture the main requirements and constraints for document 0000_cctns.

[SYSTEM] Okay, that helps clarify the project's goal. To get a better sense of the system's core functions, could you walk me through some of the most common or frequent tasks a typical user would perform in the existing cctns system?

[USER] Sure. A lot of what our users do is repetitive. For example, creating a new case or updating an existing one are things they do all day. It's really important that these frequently-executed system transactions are designed to be quick. We need people to be able to complete them with a small number of interactions, not have to click through a bunch of screens for something they do constantly.

[SYSTEM] That's a great point about minimizing clicks for frequent tasks. On that note, when users are looking at a complex case file with a lot of information on one page, how should we present that data so they can easily find the specific section or link they need without a lot of scrolling or hunting around?

[USER] That's a great point. First, we need to make sure the length of a page is appropriate for what the user is trying to do. For those really long pages, we could use within-page links to help people jump to the right section, but it's critical that those links are clearly distinguishable from links that go to a different page. For instance, maybe the within-page links could have dashed rather than solid underlines. We also need to be careful that if a page has a lot of links, the text is still easy to read. And no matter where a user is, each page should always have a clear link leading back to the home page or to a landmark page they can easily recognize, so they never feel lost.

[SYSTEM] That's a great suggestion to use different underline styles for links. When we think about other visual cues, like using color to highlight important information or status, how can we make sure that meaning is also clear to someone who might be colorblind?

[USER] That's a really important point. We absolutely can't rely on color as the only means of conveying information, since we have to account for how different people perceive it. For our general approach, we should be using ISO 9241-20 as the standard for guidance on ICT accessibility. This also means that all non-text media objects, like graphical images or video, need to have equivalent textual descriptions. And to give users more control, customization should include the ability to change things like the menu contents.

[SYSTEM] You've brought up some great points about user control and customization, like changing menu contents. To build on that idea of making text easy to read for everyone, how should the system support users who might need to adjust the font size to see the content more clearly?

[USER] That's a really important point. Users absolutely need to be able to resize the text, and they should be able to do it using the functions their browser already provides or other appropriate means. Thinking about general usability, we should also avoid splash screens unless they're providing some kind of useful content or feedback on the application state. If we do end up using a splash screen, we have to offer a navigation option to skip it. And one more thing on this topic – if the team is considering using frames, we need to be very careful to avoid problems we've seen before, like issues with the back button, bookmarking pages, or weird scrolling behavior.

[SYSTEM] You've given some great examples of how to make the system visually accessible. To extend that, how should we ensure that someone who can't use a mouse, and relies solely on a keyboard or another input device, can still perform all the necessary tasks efficiently?

[USER] That's a crucial point. The user interfaces should be designed to allow activation of controls by a variety of input devices, not just a mouse. To help people who can't easily browse, we should also provide a site map that shows the structure of the site in an overview form. It's also important that if we have navigation components on the screen, they stay consistent with the content being shown. For example, the navigation should always indicate the user's current location. Finally, if the system is ever using individualization and/or adaptation for someone, it should be made evident to the user so they understand why things might look different.

[SYSTEM] You've given some great ideas for helping users navigate, like providing a site map and always showing their current location. Building on that, if someone is in the middle of a task and isn't sure what to do next, how should the system provide help or instructions to guide them?

[USER] Yes, that's a really important point. If a user gets stuck, we need to provide help right there on the screen. It should be context-sensitive, meaning the help material is specific to whatever they're trying to do in that part of the application. This needs to be available for all the possible actions and scenarios on all user interfaces. As for the help content itself, it has to be easy to digest. We should use things like short sentences, break up long paragraphs into shorter chunks, and maybe use bullet points. And of course, the quality of the writing matters—it needs to have good spelling and grammar so it's clear and doesn't hurt readability.

[SYSTEM] That's a great point about making help content easy to digest using formatting like short paragraphs. To extend that idea of readability to the entire interface, how should we approach the overall page layout to help users quickly scan and find what they're looking for?

[USER] That's a great point. The overall design should really take into account the general principles of human perception so it feels natural to use. For instance, page titles should always be placed in a consistent location on every page so users know where they are. We also need to be smart about how we use empty space. Any space filled only with the background color should be used in a way that it doesn't impair the visual skimming of the page. And thinking bigger, we need to minimize the number of navigation steps it takes to get to a piece of content, but we have to do that while considering that different users have different mental models, navigation strategies, and tasks they're trying to complete.

[SYSTEM] You brought up a great point about keeping page titles in a consistent location. To expand on that idea of consistency, how should we approach the overall design to ensure that elements like buttons, links, and menus have a uniform appearance and behave predictably across the entire application?

[USER] That's a great question. To make it feel familiar and predictable, we really need to use a single set of user interface rules to provide a common look and feel across the entire application. It's also critical that the user interface is suitable for users with special needs, so we're not excluding anyone. On a practical note, something that really frustrates users is horizontal scrolling, so we should avoid that wherever possible. And for navigation, it's important that we use familiar terminology for the links so people know exactly where they're going without having to learn a new vocabulary.

[SYSTEM] You made an excellent point about making the interface suitable for users with special needs. To build on that, what should happen if a user's browser or assistive technology can't render our standard visual styles? How can we ensure the content is still perfectly readable and easy to navigate in a simplified or text-only format?

[USER] That's exactly the kind of scenario we need to plan for. If a user has style sheets or frames turned off, it's critical that they can still read and understand the page without any issues. We need to ensure the text is always readable and the spatial arrangement makes sense, even in a simplified view. Providing an equivalent alternative text-only page seems like a good solution to make sure the content is always accessible. Ultimately, we should be following the established best practices for this; our user interfaces should follow the guidelines specified under www.usability.gov to make sure we're meeting those standards.

[SYSTEM] You've brought up several great points about navigation, like using familiar terms for links. To make sure users always know what to expect, how should we visually differentiate a link that takes them to another page from a button that performs an action, like submitting a form?

[USER] Yes, that's a really important point. Users need to know what to expect, so navigation links should be clearly distinguishable from controls that activate some action. It's also easy for people to get lost, so every page or window should provide a clear indication of where the user is in the navigation structure. We also need to support different navigation strategies by offering alternative access paths to get to the same content. And one more thing on links – if a link is going to open new browser windows or pop-up windows, it absolutely needs to be clearly marked so people aren't surprised.

[SYSTEM] You mentioned providing alternative access paths to content, which is a great point for flexibility. Applying that idea to our help and support system, should users be able to find answers both from within the application and also through a separate, standalone support website?

[USER] Yes, absolutely. We need users to be able to find support both from within the application and also through a separate browser interface. It's also important that the help content is easy for everyone to access. For instance, if frames are used for the presentation, we must provide an alternative way of presenting the information without frames, as that can be a big usability issue.

[SYSTEM] That's very clear on the help system. Shifting gears a little to system administration and security, if an external auditor ever needed to review user activity for a specific case, how should we provide them with that information while also guaranteeing that the original log in our system remains completely secure and unchangeable?

[USER] That's a really important point. We need to be able to export audit trails for specified cases for an auditor, but doing so can't have any effect on the audit trail stored by the System. That original log is our source of truth and must be completely unalterable, meaning the audit trail data cannot be modified in any way or deleted by any user. The log also needs to be comprehensive, so if the System allows users to make unauthorised attempts to access cases, it must log these in the audit trail. As for how long we keep them, the system must maintain the audit trail for at least for the life of the case to which it refers.

[SYSTEM] That's very clear on protecting the audit trail. When the system does log a security violation, like an unauthorized access attempt, what should happen next? For instance, should an administrator be notified immediately?

[USER] Yes, the system absolutely must be able to capture and store violations. To help prevent those, it's critical that changes to security attributes for groups or users can be made only by a super-user. That includes things like access rights, security level, privileges, and password allocation and management. On the other hand, if we use user-specified profiles, the users themselves should be able to see, modify, and delete that profile on demand. And while we're on the topic of user feedback, we should also provide an interface for anyone to log any defects or enhancement requests on the application and be able to track them afterwards.

[SYSTEM] You mentioned users should be able to modify their own profiles. To help make their day-to-day work more efficient, should they also be able to set up their own persistent default values for common data entry fields?

[USER] Yes, absolutely. Letting users set up their own persistent defaults for data entry would be a great efficiency boost, especially since they should be able to set their own user-definable values. And while we're on the topic of data entry, we need to make sure that when something goes wrong, the error messages are actually helpful. They must be meaningful to the people who see them. A good error message should clearly state the reason for the problem and tell the user what actions they can take to resolve the error.

[SYSTEM] That's a great point about making error messages helpful. Thinking about the initial data entry, for example when a citizen is registering a new complaint, what kind of checks should the system run to ensure the information is valid and complete before it's accepted?

[USER] Well, the main goal is for citizens to be able to register their complaints so the police can investigate them properly. To make sure the data is complete, it would be helpful if we could set up some default values for certain fields, and it's important that those are user-definable values so we can control them. For the specific validation rules, we should probably consult the relevant International Standards for guidance to make sure we're aligned with best practices.

[SYSTEM] That makes sense to align with international standards for the data itself. Once a complaint is successfully registered, what other systems, like police case management software, would need to receive this information to help with the investigation?

[USER] That's a good point. Generally speaking, any interface we build needs to be easy to use and intuitive, not just for our end users but for administrators as well. We'd also want the interfaces to be user-configurable to the extent possible. It's also really important that the system supports a multilingual interface for our users. And from a reliability standpoint, it's critical that the system is designed so that operational data is not lost in case of any failure of equipment or the communication network. It should also be compatible with any specialist software we might use down the line.

[SYSTEM] You mentioned reliability, which is crucial. To make sure the system also feels responsive in day-to-day work, what's the longest an officer should have to wait for a case file to open, whether it's a recent one or an older one from the archives?

[USER] That's a key point. We need acceptable opening times across the board. For a recent case, one that's been accessed within the previous 2 months, an officer should see it on screen within 5-8 seconds. For older cases that haven't been accessed in that time, we understand it might take longer to pull from the archives, so up to 20 seconds is fine. These times are important and should hold up regardless of the storage capacity or the total number of cases in the system. Also, officers often need to compare information, so the system needs to be able to display several entities, like cases and suspects, simultaneously without bogging down.

[SYSTEM] Those response time targets are very helpful. On a related note, thinking about times the system might be unavailable for maintenance or due to unexpected issues, how much total downtime would be considered acceptable over a typical three-month period?

[USER] That's a great question. Reliability is key for us. For planned maintenance, we can't really have more than <xx> hours of downtime in any rolling three month period. When it comes to unplanned outages, we need to be even stricter. We can't have more than <x> incidents of unplanned downtime in that same three-month window, and the total time it's down can't add up to more than <xx hours/minutes>. And if there's a major failure, like a software or hardware issue, the absolute priority is getting it back online. We need the system to be fully restored, including with inline synchronization, in no more than <xx> hours.

[SYSTEM] We've established clear targets for system availability, which is great for ensuring officers can do their work. To make sure they can do their work securely, what should the system show an officer if they happen to search for a case that's outside their authorized access level?

[USER] That's a critical question for us. If a user performs any kind of search, the system must never include a record in the search result list if the user doesn't have the right to access it. For situations where a user tries to access a specific case they don't have rights to, we need some flexibility. The system should let us choose what happens at configuration time; for instance, one option could be to just 'display title and metadata'. This all ties into a larger need for role-based control for functionality, and it's important that a user can be a member of more than one group. We also need to allow a user to stipulate which other users or groups can access their cases.

Part C - Extracted requirements by category

Functional

FR-001	The system shall allow a user to create a new case.
FR-002	The system shall allow a user to update an existing case.
FR-003	For long pages, the system shall provide within-page links to help users jump to the right section.
FR-004	Each page must always have a clear link leading back to a landmark page they can easily recognize or to the home page.
FR-005	Users must have the ability to change the menu contents.
FR-006	Users must be able to resize text.
FR-007	If a splash screen is used, it must provide a navigation option to skip it.
FR-008	The system must provide a site map that shows the structure of the site in an overview form.
FR-009	The navigation must always indicate the user's current location.

FR-010 If the system is using individualization and/or adaptation for a user, it must be made evident to that user.

FR-011 The system must provide help directly on the screen.

FR-012 Help material must be context-sensitive to the user's current task within the application.

FR-013 Context-sensitive help must be available for all possible actions and scenarios on all user interfaces.

FR-014 The system must provide an equivalent alternative text-only page to ensure content is always accessible.

FR-015 Every page or window must provide a clear indication of the user's location in the navigation structure.

FR-016 The system must support different navigation strategies by offering alternative access paths to the same content.

FR-017 Users must be able to find support from within the application.

FR-018 If frames are used for the presentation of help content, an alternative way of presenting the information without frames must be provided.

FR-019 The system must be able to export audit trails for specified cases for an auditor.

FR-020 Users must be able to see their user-specified profile on demand.

FR-021 Users must be able to modify their user-specified profile on demand.

FR-022 Users must be able to delete their user-specified profile on demand.

FR-023 Users must be able to track the defects or enhancement requests they have logged.

FR-024 Users must be able to set up their own persistent defaults for data entry.

FR-025 Citizens must be able to register their complaints.

FR-026 Interfaces must be user-configurable to the extent possible.

FR-027 The system must support a multilingual interface.

FR-028 The system must be able to display several entities, like cases and suspects, simultaneously.

FR-029 The system's response when a user tries to access a specific case they do not have rights to must be configurable at configuration time.

FR-030 A configurable option for when a user tries to access a specific case they do not have rights to is to 'display title and metadata'.

FR-031 A user must be able to be a member of more than one group.

FR-032 A user must be able to stipulate which other users or user groups can access their cases.

Non-Functional

NFR-001 Frequently-executed system transactions must be designed to be quick.

NFR-002 Users must be able to complete frequently-executed system transactions with a small number of interactions.

NFR-003 The length of a page must be appropriate for what the user is trying to do.

NFR-004 Within-page links must be clearly distinguishable from links that go to a different page.

NFR-005 Within-page links could have dashed rather than solid underlines to distinguish them.

NFR-006 If a page has a lot of links, the text must still be easy to read.

NFR-007 Color must not be the only means of conveying information.

NFR-008 The system must use ISO 9241-20 as the standard for guidance on ICT accessibility.

NFR-009 All non-text media objects, like graphical images or video, must have equivalent textual descriptions.

NFR-010 The system must avoid using splash screens unless they provide useful content or feedback on the application state.

NFR-011 If frames are used, they must not cause issues with the back button.

NFR-012 If frames are used, they must not cause issues with bookmarking pages.

NFR-013 If frames are used, they must not cause weird scrolling behavior.

NFR-014 User interfaces must allow activation of controls by a variety of input devices, not just a mouse.

NFR-015 Navigation components on the screen must remain consistent with the content being shown.

NFR-016 Help content must be easy to digest, using techniques like short sentences, shorter paragraphs, and bullet points.

NFR-017 Help content must have good spelling and grammar.

NFR-018 The overall design must take into account the general principles of human perception.

NFR-019 Page titles should always be placed in a consistent location on every page.

NFR-020 Empty space filled only with the background color must be used in a way that does not impair the visual skimming of the page.

NFR-021 The system must minimize the number of navigation steps required to get to a piece of content, while considering that different users have different mental models, navigation strategies, and tasks they're trying to complete.

NFR-022 A single set of user interface rules must be used to provide a common look and feel across the entire application.

NFR-023 The user interface must be suitable for users with special needs.

NFR-024 The system should avoid horizontal scrolling wherever possible.

NFR-025 Navigation links must use familiar terminology.

NFR-026 If a user has style sheets or frames turned off, they must still be able to read and understand the page.

NFR-027 The text must always be readable and the spatial arrangement must make sense, even in a simplified view.

NFR-028 User interfaces must follow the guidelines specified under www.usability.gov.

NFR-029 Navigation links must be clearly distinguishable from controls that activate an action.

NFR-030 Links that open new browser windows or pop-up windows must be clearly marked.

NFR-031 The audit trail data stored by the system cannot be modified in any way or deleted by any user.

NFR-032 If the System allows users to make unauthorised attempts to access cases, it must log these in the audit trail.

NFR-033 The system must be able to capture and store security violations.

NFR-034 Error messages must be meaningful to the people who see them.

NFR-035 Error messages must clearly state the reason for the problem.

NFR-036 Error messages must tell the user what actions they can take to resolve the error.

NFR-037 Data validation rules should consult the relevant International Standards for guidance.

NFR-038 Interfaces must be easy to use and intuitive for end users and administrators.

NFR-039 Operational data must not be lost in case of any failure of equipment or the communication network.

NFR-040 The system must be compatible with any specialist software that might be used in the future.

NFR-041 An older case that has not been accessed within the previous 2 months must be displayed on screen in up to 20 seconds.

NFR-042 Case opening times must hold up regardless of the storage capacity or the total number of cases in the system.

NFR-043 Planned maintenance downtime must not exceed <xx> hours in any rolling three month period.

NFR-044	There must be no more than <x> incidents of unplanned downtime in any rolling three-month window.
NFR-045	The total unplanned downtime cannot add up to more than <xx hours/minutes> in any rolling three-month window.
NFR-046	After a major failure, the system must be fully restored, including with inline synchronization, in no more than <xx> hours.
NFR-047	The system must provide role-based control for functionality.
NFR-048	The writing needs to have good spelling and grammar so it's clear and doesn't hurt readability.

Data

DATA-001	The system must maintain the audit trail for at least the life of the case to which it refers.
DATA-002	Security attributes include access rights, security level, privileges, and password allocation and management.
DATA-003	Persistent default values for data entry must be user-definable.
DATA-004	The system must allow setting up default values for certain fields.
DATA-005	Default values for fields must be user-definable.

Business Rules

BR-001	Exporting an audit trail must have no effect on the audit trail stored by the System.
BR-002	Changes to security attributes for groups or users can be made only by a super-user.
BR-003	The system must never include a record in a search result list if the user does not have the right to access it.

Interfaces

IF-001	Users should be able to resize text using their browser's built-in functions or other appropriate means.
IF-002	Users must be able to find support through a separate browser interface.
IF-003	The system must provide an interface for anyone to log any defects or enhancement requests on the application.

Part D - Matched vs missed table

Source requirement text	Matched?	Closest generated requirement text
Registration Citizens can register their complaints with police and then based on the evidence, facts and following investigation, police shall take the complaint forward.	✓	Citizens must be able to register their complaints.
Help Module The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	✓	Help material must be context-sensitive to the user's current task within the application.
The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	✓	Context-sensitive help must be available for all possible actions and scenarios on all user interfaces.
The solution should provide an interface for the user to log any defects or enhancement requests on the application and track thereafter.	✓	Users must be able to track the defects or enhancement requests they have logged.

Source requirement text	Matched?	Closest generated requirement text
The solution should send alerts (e.g., email, SMS) to the user if the user chooses to whenever any action has been taken on the alert.	✗	— not recovered
The solution should enable the user to track the submitted defect or enhancement request.	✗	— not recovered
The solution should enable the help-desk user to view the reports on the submitted defects or enhancement requests category-wise, status-wise, and age- wise.	✗	— not recovered
The support solution should be accessible to the users both from within the application and also outside the application through a browser interface.	✓	Users must be able to find support from within the application.
The System must keep an unalterable audit trail capable of automatically capturing and storing information about: All the actions (create/read/update/delete) that are taken upon the critical entities (case, suspect, property,...) in the system The user initiating and or carrying out the action;	✗	— not recovered
Administrative parameters The word “unalterable” is to mean that the audit trail data cannot be modified in any way or deleted by any user;	✓	The audit trail data stored by the system cannot be modified in any way or deleted by any user.
it may be subject to re-department and copying to removable media if required, so long as its contents remain unchanged.	✗	— not recovered
Once the audit trail functionality has been activated, the System must track events without manual intervention, and store in the audit trail information about them.	✗	— not recovered
The System must maintain the audit trail for as long as required, which will be at least for the life of the case to which it refers.	✓	The system must maintain the audit trail for at least the life of the case to which it refers.
The System must be able to export audit trails for specified cases (without affecting the audit trail stored by the System).	✓	The system must be able to export audit trails for specified cases for an auditor.
The System must be able to capture and store violations (i.e.	✓	The system must be able to capture and store security violations.
The System must at a minimum be able to provide reports for actions on cases organised: By case;	✗	— not recovered
The System should be able to provide reports for actions on cases organised by workstation and (where technically appropriate) by network address.	✗	— not recovered
The System must allow the user to limit access to cases to specified users or user groups.	✓	The system's response when a user tries to access a specific case they do not have rights to must be configurable at configuration time.
The system should provide for role-based control for the functionality within the system The System must allow a user to be a member of more than one group.	✓	The system must provide role-based control for functionality.
The System must allow only admin-users to set up user profiles and allocate users to groups.	✓	A user must be able to be a member of more than one group.
The System should allow a user to stipulate which other users or groups can access cases.	✓	A user must be able to stipulate which other users or user groups can access their cases.

Source requirement text	Matched?	Closest generated requirement text
The System must allow changes to security attributes for groups or users (such as access rights, security level, privileges, password allocation and management) to be made only by super-user.	✓	Changes to security attributes for groups or users can be made only by a super-user.
If a user requests access to, or searches for, a case which he does not have the right to access, the System must provide one of the following responses (selectable at configuration time): display title and metadata;	✓	A configurable option for when a user tries to access a specific case they do not have rights to is to 'display title and metadata'.
the most stringent) implies that the System must not include such cases in any count of search results;	✗	— not recovered
If a user performs a quick or advanced search, the System must never include in the search result list any record which the user does not have the right to access.	✓	The system must never include a record in a search result list if the user does not have the right to access it.
If the System allows users to make unauthorised attempts to access cases, it must log these in the audit trail.	✓	If the System allows users to make unauthorised attempts to access cases, it must log these in the audit trail.
Any access to cases, and all other activities involving the cases and related documents or data should also need to be stored in the audit trail to ensure legal admissibility and to assist in data recovery.	✗	— not recovered
Ease of Use All error messages produced by the System must be meaningful, so that they can be appropriately acted upon by the users who are likely to see them.	✓	Error messages must be meaningful to the people who see them.
The System must employ a single set of user interface rules, or a small number of sets to provide a familiar and common look and feel for the application.	✓	A single set of user interface rules must be used to provide a common look and feel across the entire application.
The System must be able to display several entities (cases, suspects) simultaneously.	✓	The system must be able to display several entities, like cases and suspects, simultaneously.
The interfaces must be made customizable or user-configurable to the extent possible.	✓	Interfaces must be user-configurable to the extent possible.
Such configurations must be saved in the user profile.	✗	— not recovered
The System user interface must be suitable for users with special needs;	✓	The user interface must be suitable for users with special needs.
that is, compatible with specialist software that may be used and with appropriate interface guidelines The System must provide End User and Administrator functions which are easy to use and intuitive throughout.	✓	The system must be compatible with any specialist software that might be used in the future.
The System must allow persistent defaults for data entry where desirable.	✓	Users must be able to set up their own persistent defaults for data entry.
These defaults should include: user-definable values;	✓	Default values for fields must be user-definable.
Frequently-executed System transactions must be designed so that they can be completed with a small number of interactions (e.g.	✓	Users must be able to complete frequently-executed system transactions with a small number of interactions.
Where the System employs a graphical user interface, it must allow users to customise it.	✗	— not recovered
Customisation should include, but need not be limited to the following changes: menu contents;	✓	Users must have the ability to change the menu contents.

Source requirement text	Matched?	Closest generated requirement text
Usability The user interfaces should be designed to make them user-intuitive.	✓	Interfaces must be easy to use and intuitive for end users and administrators.
The user interfaces of the system should comply with Standard ISO 9241.	✗	— not recovered
ICT accessibility: ISO 9241-20 shall be the standard for guidance on ICT accessibility.	✓	The system must use ISO 9241-20 as the standard for guidance on ICT accessibility.
Software accessibility ISO 9241-171 shall be the standard for guidance on software accessibility.	✗	— not recovered
User interfaces should meet its requirements and recommendations.	✗	— not recovered
Content accessibility WCAG 1.0 shall be the standard used for guidance on content accessibility.	✗	— not recovered
Providing text equivalents for non-text media objects: All non-text media objects, such as graphical images or video, should be provided with alternative equivalent textual descriptions and/or with equivalent text-based functionality.	✓	All non-text media objects, like graphical images or video, must have equivalent textual descriptions.
Making navigation self-descriptive: Navigation should be designed to help users understand where they are, where they have been and where they can go next.	✓	The navigation must always indicate the user's current location.
Showing users where they are: Each presentation segment (page or window) should provide the user with a clear and sufficient indication of where he or she is in the navigation structure and of the current segment position with respect to the overall structure.	✓	Every page or window must provide a clear indication of the user's location in the navigation structure.
Offering alternative access paths: Alternative access paths for navigating to a specific unit of content should be offered to support different navigation strategies.	✓	The system must support different navigation strategies by offering alternative access paths to the same content.
Minimizing navigation effort: The number of navigation steps needed to reach a certain piece of content should be minimized as long as different mental models, navigation strategies and tasks of the user are taken into account.	✓	The system must minimize the number of navigation steps required to get to a piece of content, while considering that different users have different mental models, navigation strategies, and tasks they're trying to complete.
Splash screens should be avoided unless they provide useful content or feedback about the application state to the user.	✓	The system must avoid using splash screens unless they provide useful content or feedback on the application state.
If a splash screen is used, a navigation option to skip it should be offered.	✓	If a splash screen is used, it must provide a navigation option to skip it.
Avoiding opening unnecessary windows: Additional windows such as new browser windows or pop-up windows should only be opened if this supports the user’s task.	✗	— not recovered
Horizontal scrolling should be avoided wherever possible.	✓	The system should avoid horizontal scrolling wherever possible.
Designing for input device independence: User interfaces should be designed to allow activation of controls by a variety of input devices.	✓	User interfaces must allow activation of controls by a variety of input devices, not just a mouse.
Making user interfaces robust: User interfaces should be designed to be as robust as possible in the face of changing technology.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
Acceptable opening / download times: Application pages should be designed and implemented so that there are acceptable opening times and download times for the expected range of technical contexts of use (e.g.	✗	— not recovered
Minimizing user errors: Potential user errors as well as the effort needed to recover from errors should be minimized.	✗	— not recovered
Providing clear error messages: The content of error messages shown on the pages or special error pages should clearly state the reason why the error occurred and, if possible, actions the user can take to resolve the error.	✓	Error messages must tell the user what actions they can take to resolve the error.
Making text resizable by the user: Text should be able to be resized by the user, using functions provided by the user agent or other appropriate means i.e.	✓	Users must be able to resize text.
Text quality: The quality of textual content with respect to spelling and grammar should be sufficient so as not to impede readability.	✓	The writing needs to have good spelling and grammar so it's clear and doesn't hurt readability.
Writing style: The reading and understanding of the textual content on the screen should be supported by suitable means, including the use of short sentences, the division of the text into shorter chunks or the presentation of content items in the form of bullet points.	✓	Help content must be easy to digest, using techniques like short sentences, shorter paragraphs, and bullet points.
Supporting text skimming: Fast skimming of text should be supported by the provision of clear links, bulleted lists, highlighted keywords, logical headings, and short phrases and sentences.	✗	— not recovered
Readability of text: Text presented on the pages should be readable taking into account the expected display characteristics and spatial arrangement.	✓	The text must always be readable and the spatial arrangement must make sense, even in a simplified view.
ISO 9241-303 shall be consulted for screen text legibility requirements.	✗	— not recovered
Distinguishable within-page links: Within-page links should be clearly distinguishable from other links that lead to a different page.	✓	Within-page links must be clearly distinguishable from links that go to a different page.
Within-page links are shown with dashed rather than solid underlines Avoiding link overload: Text pages containing large proportions of links should be formatted so that the presence of links does not impede the readability of the text.	✓	Within-page links could have dashed rather than solid underlines to distinguish them.
Using familiar terminology for navigation links: Navigation links — particularly links representing the main navigation structure — should be labelled with terms that are familiar to the user, based on his/her general knowledge, prior experience in the application domain or experience of using other systems.	✓	Navigation links must use familiar terminology.
Marking links opening new windows: Links that open new browser windows or pop-up windows should be clearly marked.	✓	Links that open new browser windows or pop-up windows must be clearly marked.
Distinguishing navigation links from controls: Navigation links should be clearly distinguishable from controls activating some action.	✓	Navigation links must be clearly distinguishable from controls that activate an action.

Source requirement text	Matched?	Closest generated requirement text
space filled only with the background color should be used in such a way that it does not impair the visual skimming of the page.	✓	Empty space filled only with the background color must be used in a way that does not impair the visual skimming of the page.
Selecting appropriate page lengths The length of a page should be selected so as to support the primary purpose and use of the page.	✓	The length of a page must be appropriate for what the user is trying to do.
Using colour: Colour should be used with care, taking into account human capabilities and restrictions in perceiving colour, and not as the only means of conveying information.	✓	Color must not be the only means of conveying information.
Color should never be the only means of coding.	✗	— not recovered
Using frames with care: If frames are used, care should be taken to avoid possible problems, for example, those involving the use of the back button, bookmarking of pages, or scrolling of information.	✓	If frames are used, they must not cause issues with the back button.
Providing alternatives to frame-based presentation: If frames are used, an alternative way of presenting relevant information without frames should be provided.	✓	If frames are used for the presentation of help content, an alternative way of presenting the information without frames must be provided.
Providing alternative text-only pages: When style sheets and/or frames are turned off it should be possible for the user to read and understand the page;	✓	If a user has style sheets or frames turned off, they must still be able to read and understand the page.
alternatively, the user should be provided with an equivalent alternative text-only page.	✓	The system must provide an equivalent alternative text-only page to ensure content is always accessible.
Consistent page layout: Pages should be designed using consistent layout schemes, supporting the user in finding similar information at the same position on different pages.	✗	— not recovered
Placing title information consistently: Page titles should be placed in a consistent location on the different pages.	✓	Page titles should always be placed in a consistent location on every page.
Observing principles of human perception When designing application pages, the general principles of human perception should be taken into account.	✓	The overall design must take into account the general principles of human perception.
The International Standards mentioned below shall be consulted for guidance.	✓	Data validation rules should consult the relevant International Standards for guidance.
In addition, when designing multimedia information presentations, the design principles and recommendations described in ISO 14915-1 to ISO 14915-3 should be taken into account.	✗	— not recovered
Linking back to the home page or landmark pages: Each page should contain a link leading to the home page of the application or to a landmark page that is easy to recognize for the user.	✓	Each page must always have a clear link leading back to a landmark page they can easily recognize or to the home page.
Providing a site map: A separate navigation overview such as a site map should be provided for application showing the structure of the site in an overview form.	✓	The system must provide a site map that shows the structure of the site in an overview form.
Consistency between navigation components and content: If navigation components (or overviews) are shown in conjunction with associated content, consistency between the navigation component and the content shown should be maintained by indicating in the navigation component (e.g.	✓	Navigation components on the screen must remain consistent with the content being shown.

Source requirement text	Matched?	Closest generated requirement text
Placing navigation components consistently: Navigation components should be placed consistently on the pages or in the framesets in the pages of the application.	✗	— not recovered
Taking account of the users' tasks and information needs: When providing different access paths or navigation structures for different user groups, the tasks and information needs of these user groups should be taken into consideration.	✗	— not recovered
Making individualization and adaptation evident: It should be made evident to the user when individualization and/or adaptation are used.	✓	If the system is using individualization and/or adaptation for a user, it must be made evident to that user.
Making user profiles evident: If predefined user profiles or user-specified profiles are used for individualizing or adapting content, the profile currently used should be made evident.	✓	Users must be able to see their user-specified profile on demand.
Allowing users to see and change profiles: If user-specified profiles are used, users should be able to see, modify and delete that profile on demand.	✓	Users must be able to modify their user-specified profile on demand.
The user interfaces of the system should follow the guidelines specified under www.usability.gov System Availability The System must be available to users: from <xx:00> to <xx:00>;	✓	User interfaces must follow the guidelines specified under www.usability.gov.
The planned downtime for the System must not exceed <xx> hours per <rolling three month period>.	✓	Planned maintenance downtime must not exceed <xx> hours in any rolling three month period.
Unplanned downtime for the System must not exceed <xx hours/minutes> per <rolling three month period>.	✓	The total unplanned downtime cannot add up to more than <xx hours/minutes> in any rolling three-month window.
The number of incidents of unplanned downtime for the System must not exceed <x> per <rolling three month period>.	✓	There must be no more than <x> incidents of unplanned downtime in any rolling three-month window.
In the event of any software or hardware failure, it must be possible to restore the System (with inline synchronization) within no more than <xx> hours.	✓	After a major failure, the system must be fully restored, including with inline synchronization, in no more than <xx> hours.
The System must be able to retrieve and display within 5-8 seconds the case which has been accessed within the previous 2 months, regardless of storage capacity or number of cases in the system.	✓	Case opening times must hold up regardless of the storage capacity or the total number of cases in the system.
The System must be able to retrieve and display within 20 seconds the case which has not been accessed within the previous 2 months, regardless of storage capacity or number of cases in the system.	✓	An older case that has not been accessed within the previous 2 months must be displayed on screen in up to 20 seconds.
The System be scaleable and must not have any features which would preclude use in small or large police stations, with varying numbers of cases handled.	✗	— not recovered
The systems should be designed with the following broad guidelines: System Functionality The system should support multilingual interface The system should be designed in manner that operational data is not lost in case of any failure of equipment or communication network.	✓	The system must support a multilingual interface.

Part E - Gemini judge requirement matching

Strict P/R/F1: 0.4835 / 0.4400 / 0.4607. Weighted P/R/F1: 0.6264 / 0.5700 / 0.5969.

Judge scoring: strict counts only full matches. Weighted counts full=1 and partial=0.5.

Verdict	Source requirement	Generated requirement	Judge reason
full	If the System allows users to make unauthorised attempts to access cases, it must log these in the audit trail.	If the System allows users to make unauthorised attempts to access cases, it must log these in the audit trail.	The generated requirement is an exact match of the source requirement.
full	The System must be able to display several entities (cases, suspects) simultaneously.	The system must be able to display several entities, like cases and suspects, simultaneously.	The generated requirement faithfully covers the same core need and preserves all critical details, merely rephrasing the parenthetical example.
full	space filled only with the background color should be used in such a way that it does not impair the visual skimming of the page.	Empty space filled only with the background color must be used in a way that does not impair the visual skimming of the page.	The generated requirement faithfully covers the source, preserving all critical details. "Empty" clarifies "space filled only with the background color" and the modality change is a strengthening.
full	The System user interface must be suitable for users with special needs;	The user interface must be suitable for users with special needs.	The generated requirement faithfully covers the source requirement, preserving all critical details. The omission of 'System' is not a material loss as 'user interface' implies it's part of the system.
full	Marking links opening new windows: Links that open new browser windows or pop-up windows should be clearly marked.	Links that open new browser windows or pop-up windows must be clearly marked.	The generated requirement faithfully covers the source, preserving all critical details. The modality is strengthened from 'should' to 'must' but the core intent remains.
full	ICT accessibility: ISO 9241-20 shall be the standard for guidance on ICT accessibility.	The system must use ISO 9241-20 as the standard for guidance on ICT accessibility.	The generated requirement faithfully covers the source, preserving all critical details regarding the standard and its purpose.
full	If a splash screen is used, a navigation option to skip it should be offered.	If a splash screen is used, it must provide a navigation option to skip it.	The generated requirement faithfully covers the source, preserving all critical details and conditions regarding offering a skip option for splash screens.
full	The System must maintain the audit trail for as long as required, which will be at least for the life of the case to which it refers.	The system must maintain the audit trail for at least the life of the case to which it refers.	The generated requirement accurately captures the minimum retention period specified in the source.
full	If a user performs a quick or advanced search, the System must never include in the search result list any record which the user does not have the right to access.	The system must never include a record in a search result list if the user does not have the right to access it.	The generated requirement fully preserves the core need and critical details, stating that unauthorized records must never be included in search results. The generalization from 'quick or advanced search' to 'a search result list' does not lose material detail.
full	The planned downtime for the System must not exceed <xx> hours per <rolling three month period>.	Planned maintenance downtime must not exceed <xx> hours in any rolling three month period.	The generated requirement faithfully covers the source requirement regarding planned downtime limits, with 'maintenance' being a reasonable specification of 'planned'.

Verdict	Source requirement	Generated requirement	Judge reason
full	Distinguishable within-page links: Within-page links should be clearly distinguishable from other links that lead to a different page.	Within-page links must be clearly distinguishable from links that go to a different page.	The generated requirement faithfully covers the same need to distinguish within-page links from other links. The change in modality from 'should' to 'must' does not alter the core requirement.
full	Splash screens should be avoided unless they provide useful content or feedback about the application state to the user.	The system must avoid using splash screens unless they provide useful content or feedback on the application state.	The generated requirement accurately reflects the source, preserving the condition for avoiding splash screens and their exceptions.
full	Offering alternative access paths: Alternative access paths for navigating to a specific unit of content should be offered to support different navigation strategies.	The system must support different navigation strategies by offering alternative access paths to the same content.	The generated requirement faithfully covers the source, preserving all critical details about offering alternative access paths to support different navigation strategies for content.
full	The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	Context-sensitive help must be available for all possible actions and scenarios on all user interfaces.	The generated requirement faithfully covers the availability and scope of context-sensitive help, preserving all critical details.
full	Distinguishing navigation links from controls: Navigation links should be clearly distinguishable from controls activating some action.	Navigation links must be clearly distinguishable from controls that activate an action.	The generated requirement faithfully covers the source, preserving all critical details. The change in modality from "should" to "must" is a strengthening but does not alter the core requirement.

Additional partial-match examples

Source requirement	Generated requirement	Judge reason
The solution should provide detailed context-sensitive help material for all the possible actions and scenarios on all user interfaces in the application.	Help material must be context-sensitive to the user's current task within the application.	The generated requirement loses the critical detail that context-sensitive help must be available for *all* possible actions and scenarios on *all* user interfaces.
The solution should provide an interface for the user to log any defects or enhancement requests on the application and track thereafter.	The system must provide an interface for anyone to log any defects or enhancement requests on the application.	The generated requirement generalizes the role from 'the user' to 'anyone' and loses the critical detail that users should be able to track their logged requests.
The solution should enable the help-desk user to view the reports on the submitted defects or enhancement requests category-wise, status-wise, and age- wise.	Users must be able to track the defects or enhancement requests they have logged.	The generated requirement generalizes the user role and loses critical details about viewing reports category-wise, status-wise, and age-wise, only covering the general ability to track.
The support solution should be accessible to the users both from within the application and also outside the application through a browser interface.	Users must be able to find support from within the application.	The generated requirement only covers accessibility from within the application and omits the critical detail of accessibility outside the application through a browser interface.
The System must be able to export audit trails for specified cases (without affecting the audit trail stored by the System).	The system must be able to export audit trails for specified cases for an auditor.	The generated requirement adds a specific role ('for an auditor') and omits the critical detail about not affecting the stored audit trail.

Source requirement	Generated requirement	Judge reason
The System must be able to export audit trails for specified cases (without affecting the audit trail stored by the System).	The audit trail data stored by the system cannot be modified in any way or deleted by any user.	The generated requirement generalizes the condition 'without affecting the audit trail stored by the System' to a broader statement about audit trail data immutability and undeletability by any user, which is a material addition and change in scope.
The System should be able to provide reports for actions on cases organised by workstation and (where technically appropriate) by network address.	The system must be able to export audit trails for specified cases for an auditor.	The generated requirement introduces exporting audit trails for an auditor and loses the specific organization criteria (workstation, network address) mentioned in the source.
The system should provide for role-based control for the functionality within the system The System must allow a user to be a member of more than one group.	A user must be able to be a member of more than one group.	The generated requirement covers only the multi-group membership aspect and omits the detail about role-based control.
The System must allow only admin-users to set up user profiles and allocate users to groups.	Changes to security attributes for groups or users can be made only by a super-user.	The generated requirement generalizes the specific actions of setting up profiles and allocating users to groups to a broader category of 'changes to security attributes', which is a material alteration of scope.
The System should allow a user to stipulate which other users or groups can access cases.	A user must be able to stipulate which other users or user groups can access their cases.	The generated requirement changes the modality from 'should' to 'must' and specifies 'their cases' instead of general 'cases', altering the strength and scope slightly.

Document: 0000_gamma_j

Part A - Document header

Document ID	0000_gamma_j
Generation model	gemini / gemini-2.5-pro
Standard validator	gemini / gemini-2.5-flash
Dialogue method	N/A
Question algorithm	N/A
Source requirement count	57
Generated requirement count	48
Matched count	35
Clarification rounds	requested 2 / used 0
Clarification chunks	0
Semantic P / R / F1	0.7292 / 0.6140 / 0.6667
LLM strict P / R / F1	0.5000 / 0.4211 / 0.4571
LLM weighted P / R / F1	0.6354 / 0.5351 / 0.5810
Evidence / Propositions / Gap	60 / 48 / 4
Grounded after rewrite	48

Part B - Generated dialogue transcript

[SYSTEM] To start, what is the overall purpose and scope of this system?

[USER] This project is based on the 0000 - gamma j system, and the goal is to capture the main requirements and constraints for document 0000_gamma_j.

[SYSTEM] Okay, to help us start capturing those requirements for the gamma j system, could you describe the primary tasks a user will need to accomplish with it?

[USER] Well, the main thing is for customers to buy things, and for that to happen, they must have one or more items in the shopping cart. We also need to make sure the payment side works, so it's important that the system can run test credit card transactions. Looking ahead, it's also critical that Yoggie coordinates with us on future enhancements and features. And just so we're on the same page, when we talk about what it needs to do, we're focused on its required functionality—the specific behaviors the system will exhibit under certain conditions.

[SYSTEM] That makes sense. Thinking about what happens right after a successful purchase, does the system need to send out any confirmation emails to the customer or perhaps notify the vendor?

[USER] Yes, absolutely. After a purchase, we need to email both the customer and the vendor. It's really important that this happens fast, so it should be able to send those emails in less than 1 second.

[SYSTEM] That's a helpful performance detail. We've covered what happens after a successful purchase, but what should the system do if a customer's payment is declined?

[USER] When a payment is declined, that's a specific condition where the system needs to have a clear function or behavior. It's important because we have to satisfy the customer's objective, so the product must possess the capability to handle that situation properly.

[SYSTEM] That makes sense. When the system handles that declined payment, how should it present that information to the customer to make sure the message is clear and easy to act on?

[USER] To make sure the message is clear, we need to make sure we utilize consistent symbols and colors for our notifications. That way, users will immediately recognize what kind of message they're seeing, whether it's an error, a warning, or a success.

[SYSTEM] That's a great point about making notifications clear for users. Thinking about system-level events, what kind of alerts or automatic responses should happen if we detect a potential security threat, like an unusually high volume of traffic from a single source?

[USER] That's a critical area for us. If we see something like an IP DOS attack, the system absolutely needs to auto detect it and block that IP automatically. We also need it to be smart enough to detect consecutive failed login attempts, since that's another common attack vector. For overall protection, we're planning to have the system protected by that open source firewall called Firestarter. And of course, it's a given that all sensitive information must be encrypted via https.

[SYSTEM] That's very clear on the security front. Thinking about the data itself—like the inventory details and customer information that needs to be protected—do you have any specific requirements for the database system where all this will be stored?

[USER] Yes, we do. It's a firm constraint that we must use a SQL based database for this project, and that's where all the inventory information will need to be stored. Beyond that, we have a couple of other data integrity concerns. When users are adding product photos, we need the system to automatically compress image files that are too large in size. And for customer data, it's really important that the system can validate email address existence when a new user signs up, to make sure we're getting legitimate contact information.

[SYSTEM] That's great information on validating emails and compressing images. You also mentioned the need for a SQL database to store this data. When we're storing sensitive customer information in that database, should we be taking any specific measures, like encryption, to protect it?

[USER] Yes, absolutely. Protecting our customers' information is a top priority for us. We need to encrypt all customer data in the database. And you're correct, we've decided we must use a SQL based database for this.

[SYSTEM] Thanks for clarifying the encryption requirement. Since the system will be built on a SQL database, is there a particular one, like PostgreSQL or MySQL, that your team already uses or prefers?

[USER] We don't have a strong preference for a specific one like PostgreSQL or MySQL. The main thing for us is that it must use a SQL based database. As long as that requirement is met, we'll leave the final choice up to the technical team.

[SYSTEM] That's helpful regarding the database choice. Thinking about how the system communicates with customers after a purchase, what kind of information should we automatically include in the order confirmation email?

[USER] The confirmation email should definitely tell the customer that their items will be delivered to their shipping address and it absolutely must include the tracking number. On a related note, if any customers or our own sales people have technical queries, we need an e-mail to go straight to the System Administrator. This is all part of making sure the experience is smooth for the customer, so the interface has to be intuitive and remain consistent among various web browsers. And just to be clear, the whole Web Store needs to operate on our required platform, which is an Intel based system with Slackware Linux 2.6 and an Apache Web Server.

[SYSTEM] That's a helpful detail about routing technical queries. To make sure we handle that correctly, how would a customer or salesperson initiate that email to the System Administrator—is it through a specific 'contact support' form on the website?

[USER] That's a good question, but I'm more concerned about the communications that happen after an order is placed. When a customer uses the WebOrder system, we need to send an e-mail confirmation letting them know the items they ordered will be delivered to their shipping address, and it's very important that this email includes the tracking number. For any of this to happen, of course, the customer must be able to access the web order system through a web browser using HTTPS for security.

[SYSTEM] You’ve emphasized the importance of a smooth experience for the customer. To ensure the site feels responsive, when a customer searches for a specific product, what's our goal for how quickly those results should appear?

[USER] A customer should be able to search for a specified product and see the results in less than 1 second. To make that happen, I know the team said we need to perform searches using Dijkstra's shortest path algorithm. Overall, the system needs to be pretty robust, able to retrieve 200 products per second to handle the load. And on a related note about performance, if something goes wrong, it's important that we can restore 1000 records per second to get back up and running quickly.

[SYSTEM] That's a great performance target for product retrieval. To get a fuller picture of the system's load, how many customers do we anticipate will be logged in and actively using the site at the same time during your busiest periods?

[USER] During our busiest periods, we need to be able to handle 1000 customers logged in concurrently at the same time. The experience for those users has to be really snappy. For instance, adding a product to the shopping cart should be incredibly fast, less than 2ms. And when a customer is ready to buy, we need to acquire the shipping charges in less than 2 seconds. We also have an operational requirement that's important: upon the USB being plugged in, the system has to be deployed and fully operational in less than 1 minute.

[SYSTEM] You mentioned that acquiring shipping charges needs to be very fast. To make that calculation possible, what information must a customer have already provided before they can see those charges?

[USER] Right, for that to work, there's a pre-condition. Basically, a certain condition must be satisfied or the system has to be in a specific state before that use case can even begin.

[SYSTEM] You've brought up several different users like customers, salespeople, and administrators. It sounds like the first step for any of them is to have an account. Once they're logged in, what are the key differences in what each of them can see and do within the system?

[USER] That's a great question. You're right, the first step for any of these users is getting set up. A Customer, for instance, has to have a registered account before they can use the Weborder System. It's the same for our Sales Person role – they also need an account to be registered. And for an Administrator, not only do they need a registered account, but they must be logged-in on the system to perform any of their functions.

[SYSTEM] It's clear that keeping the system running smoothly for our customers is a top priority. To make sure it's always there when they need it, how much total downtime, for things like maintenance or unexpected issues, would be acceptable over the course of a year?

[USER] We need this to be extremely reliable for our customers, so it should have an availability of 99.99%. And if something does go wrong, we need a simple recovery process. It should be easy to migrate the system or have it backed up via another usb drive to get things running again quickly.

[SYSTEM] You mentioned deploying and recovering the system from a USB drive, which sounds very self-contained. To help us understand the full environment, will the hardware it's running on have an internet connection for things like security patches and updates?

[USER] Yes, absolutely. The hardware will need an internet connection so it can be fixed and patched. While we're on the topic of being online, the Web Store needs to operate with some specific browsers for our users; it has to work with Microsoft Internet Explorer version 6 and 7, and also Netscape Communicator Version 4 and 5. It's also important that we're able to create a test environment of the weborder system to make sure everything is working correctly before it goes live.

Part C - Extracted requirements by category

Functional

FR-001	The system must be able to run test credit card transactions.
FR-002	The system must email both the customer and the vendor after a purchase.
FR-003	The system needs to have a clear function or behavior for when a payment is declined.
FR-004	Notifications must support error, warning, and success message types.
FR-005	The system must automatically compress image files that are too large in size.
FR-006	The order confirmation email must state that items will be delivered to the customer's shipping address.
FR-007	The order confirmation email must include the tracking number.
FR-008	The system must automatically compress image files that are too large in size when users are adding product photos.

Non-Functional

NFR-001	The system should be able to send emails in less than 1 second after a purchase.
NFR-002	The system must utilize consistent symbols and colors for notifications.
NFR-003	The system must automatically detect IP DOS attacks.

NFR-004 The system must automatically block the IP address from a detected IP DOS attack.

NFR-005 The system must detect consecutive failed login attempts.

NFR-006 All sensitive information must be encrypted via https.

NFR-007 All customer data in the database must be encrypted.

NFR-008 The interface has to be intuitive.

NFR-009 The interface must remain consistent among various web browsers.

NFR-010 A customer search for a specified product must show results in less than 1 second.

NFR-011 The system must be able to retrieve 200 products per second.

NFR-012 The system must be able to restore 1000 records per second.

NFR-013 The system must be able to handle 1000 customers logged in concurrently.

NFR-014 Adding a product to the shopping cart must take less than 2ms.

NFR-015 The system must acquire shipping charges in less than 2 seconds.

NFR-016 The system should have an availability of 99.99%.

NFR-017 The system must be able to be migrated or backed up via a USB drive.

NFR-018 It should be easy to migrate the system or have it backed up via a USB drive.

NFR-019 The system should get running again quickly after a recovery.

Data

DATA-001 All inventory information must be stored in the SQL based database.

DATA-002 The system must validate the existence of an email address when a new user signs up.

Business Rules

BR-001 A customer must have one or more items in the shopping cart to buy things.

BR-002 A Customer must have a registered account to use the Weborder System.

BR-003 A Sales Person must have a registered account.

BR-004 An Administrator must have a registered account.

BR-005 An Administrator must be logged-in on the system to perform any of their functions.

BR-006 A certain condition must be satisfied or the system has to be in a specific state before a use case can begin.

Interfaces

IF-001 An email for technical queries from customers or sales people must be sent directly to the System Administrator.

IF-002 The customer must be able to access the web order system through a web browser using HTTPS.

IF-003 The Web Store must work with Microsoft Internet Explorer version 6.

IF-004 The Web Store must work with Netscape Communicator Version 4.

Constraints

C-001 The system must be protected by the open source firewall called Firestarter.

C-002 The system must use a SQL based database.

C-003 The Web Store must operate on an Intel based system.

C-004	The Web Store must operate on Slackware Linux 2.6.
C-005	The Web Store must operate with an Apache Web Server.
C-006	Searches must be performed using Dijkstra's shortest path algorithm.
C-007	The system must be deployed and fully operational in less than 1 minute upon a USB being plugged in.
C-008	The hardware will need an internet connection for fixes and patches.
C-009	It must be possible to create a test environment of the weborder system.

Part D - Matched vs missed table

Source requirement text	Matched?	Closest generated requirement text
Operating Environment Web Store shall operate with the following internet browsers: Microsoft Internet Explorer version 6 and 7, Netscape Communicator Version 4 and 5.	✓	The Web Store must work with Microsoft Internet Explorer version 6.
Web Store shall operate on an Intel based system with Slackware Linux 2.6 and Apache Web Server.	✓	The Web Store must operate on Slackware Linux 2.6.
The system shall use SQL based database to store inventory information.	✓	All inventory information must be stored in the SQL based database.
Design and Implementation Constraints Must use a SQL based database.	✓	The system must use a SQL based database.
User Documentation Shall install online help for users via the web interface Shall deliver Operations and Maintenance manual, Users Guide book, and Installation Instructions for the Administrator.	✗	— not recovered
The interface must remain consistent among various web browsers and be intuitive to the customer.	✓	The interface must remain consistent among various web browsers.
SI-1.3: Plug-ins interface Communications Interfaces CI-1: The WebOrder system shall send an e-mail confirmation to the customer that the items they ordered will be delivered to the shipping address along with tracking number.	✗	— not recovered
CI-2: The WebOrder system shall send an e-mail to System Administrator regarding any technical queries from customers or sales people.	✓	An email for technical queries from customers or sales people must be sent directly to the System Administrator.
Quality Attribute Requirements Upon the USB being plugged in the system shall be able to be deployed and operational in less than 1 minute.	✓	The system must be deployed and fully operational in less than 1 minute upon a USB being plugged in.
The system shall be able to handle 1000 customers logged in concurrently at the same time.	✓	The system must be able to handle 1000 customers logged in concurrently.
The system shall be able to retrieve 200 products per second.	✓	The system must be able to retrieve 200 products per second.
The system shall be able to add product to shopping cart in less than 2ms.	✓	Adding a product to the shopping cart must take less than 2ms.
The system shall be able to search for a specified product in less than 1 second.	✓	A customer search for a specified product must show results in less than 1 second.
The system shall be able to email customer and vendor in less than 1 second.	✓	The system should be able to send emails in less than 1 second after a purchase.
The system shall be able to validate credit card in less than 2 seconds.	✗	— not recovered

Source requirement text	Matched?	Closest generated requirement text
The system shall be able to acquire shipping charges in less than 2 seconds.	✓	The system must acquire shipping charges in less than 2 seconds.
The system shall be able to restore 1000 records per second.	✓	The system must be able to restore 1000 records per second.
Security Requirements The system shall validate credit cards against fraud.	✗	— not recovered
The system shall encrypt all sensitive information via https.	✓	All sensitive information must be encrypted via https.
The system shall encrypt all customer data in database.	✓	All customer data in the database must be encrypted.
The system shall auto detect IP DOS attacks and block IP automatically.	✓	The system must automatically detect IP DOS attacks.
The system shall detect consecutive failed login attempts.	✓	The system must detect consecutive failed login attempts.
The system shall be protected by open source firewall called Firestarter.	✓	The system must be protected by the open source firewall called Firestarter.
http://www.fs-security.com/ Availability Requirements The system shall have an availability of 99.99%.	✓	The system should have an availability of 99.99%.
Efficiency Requirements The system shall perform searches via Dijkstra's shortest path algorithm.	✓	Searches must be performed using Dijkstra's shortest path algorithm.
For returning customers, the system shall validate 'existing' credit card in system after each log in.	✗	— not recovered
The system shall automatically compress image files that are too large in size.	✓	The system must automatically compress image files that are too large in size.
The system shall validate email address existence.	✓	The system must validate the existence of an email address when a new user signs up.
The system shall utilize consistent symbols and colors for clear notifications.	✓	The system must utilize consistent symbols and colors for notifications.
Maintainability Requirements The system shall utilize interchangeable plugins.	✗	— not recovered
The system shall be easily updatable for fixes and patches.	✗	— not recovered
The system shall create logs of all changes, updates, or fixes that are done to the site.	✗	— not recovered
The system shall be easy to upgrade.	✗	— not recovered
Portability Requirements The system shall be extremely portable via the usb drive.	✗	— not recovered
The system shall be easy to migrate or backed up via another usb drive.	✓	It should be easy to migrate the system or have it backed up via a USB drive.
Testability Requirements The system should be able to run under debug mode.	✗	— not recovered
The system should be able to run test credit card transactions.	✓	The system must be able to run test credit card transactions.
The system should be able to run test shipping orders.	✗	— not recovered
The system should be able to create test environment of weborder system.	✓	It must be possible to create a test environment of the weborder system.

Source requirement text	Matched?	Closest generated requirement text
Other Requirements The system hardware shall be fixed and patched via an internet connection.	✓	The hardware will need an internet connection for fixes and patches.
Yoggie shall coordinate on future enhancement and features with our organization.	✗	— not recovered
Functional requirements: A statement of a piece of required functionality or a behavior that a system will exhibit under specific conditions.	✗	— not recovered
Nonfunctional requirements: A description of a property or characteristic that the system should exhibit.	✗	— not recovered
Pre condition: A condition that must be satisfied or a state the system must be in before a use case may begin.	✓	A certain condition must be satisfied or the system has to be in a specific state before a use case can begin.
Requirement: A statement of a customer need or objective or of a condition or capability that a product must possess to satisfy such a need or objective.	✗	— not recovered
Actors: Customer Weborder System Preconditions: Customer must be able to access the web order system via a web browser with HTTPS.	✓	The customer must be able to access the web order system through a web browser using HTTPS.
Actors: Customer Weborder System Preconditions: Customer account must already be registered.	✗	— not recovered
Actors: Customer Weborder System Preconditions: Customer must be logged-in on the system.	✗	— not recovered
Add Item To Cart Goal Customer adds item(s) in the cart Actors: Customer Weborder System Preconditions: The customer must be logged-in on the system.	✗	— not recovered
Customer must have item(s) in the shopping cart.	✓	A customer must have one or more items in the shopping cart to buy things.
Actors: Administrator Weborder System Preconditions: Administrator account must already be registered.	✓	An Administrator must have a registered account.
Actors: Administrator Weborder System Preconditions: Administrator must be logged-in on the system.	✓	An Administrator must be logged-in on the system to perform any of their functions.
Add User Goal Register a new customer, sales person, or administrator account with the system Actors: Administrator Weborder System Preconditions: Administrator must be able to access the weborder system via a web browser with HTTPS.	✓	A Customer must have a registered account to use the Weborder System.
Actors: Administrator Weborder System Preconditions: Administrator must be able to access the weborder system via a web browser with HTTPS.	✗	— not recovered
Actors: Sales Person Weborder System Preconditions: Sales Person account must already be registered.	✓	A Sales Person must have a registered account.
Actors: Sales Person Weborder System Preconditions: Sales Person must be logged-in on the system.	✗	— not recovered
Actors: Sales Person Weborder System Preconditions: Sales Person must be able to access the weborder system via a web browser with HTTPS.	✗	— not recovered

Part E - Gemini judge requirement matching

Strict P/R/F1: 0.5000 / 0.4211 / 0.4571. Weighted P/R/F1: 0.6354 / 0.5351 / 0.5810.

Judge scoring: strict counts only full matches. Weighted counts full=1 and partial=0.5.

Verdict	Source requirement	Generated requirement	Judge reason
full	The system shall be able to retrieve 200 products per second.	The system must be able to retrieve 200 products per second.	The generated requirement faithfully covers the source, preserving all critical details.
full	The system shall be able to restore 1000 records per second.	The system must be able to restore 1000 records per second.	The generated requirement faithfully covers the source, preserving all critical details including the action, quantity, and rate.
full	The system shall detect consecutive failed login attempts.	The system must detect consecutive failed login attempts.	The generated requirement is an exact match to the source requirement.
full	The system shall be protected by open source firewall called Firestarter.	The system must be protected by the open source firewall called Firestarter.	The generated requirement is an exact match to the source requirement, preserving all critical details.
full	The system shall automatically compress image files that are too large in size.	The system must automatically compress image files that are too large in size.	The generated requirement faithfully covers the source requirement, preserving all critical details. The change from 'shall' to 'must' does not alter the mandatory nature of the requirement.
full	The system shall be able to acquire shipping charges in less than 2 seconds.	The system must acquire shipping charges in less than 2 seconds.	The generated requirement faithfully covers all aspects of the source requirement.
full	The system shall utilize consistent symbols and colors for clear notifications.	The system must utilize consistent symbols and colors for notifications.	The generated requirement faithfully covers the source, preserving the core need for consistent symbols and colors for notifications. The omission of 'clear' does not materially alter the requirement.
full	The system shall be able to handle 1000 customers logged in concurrently at the same time.	The system must be able to handle 1000 customers logged in concurrently.	The generated requirement faithfully covers the source, preserving all critical details. 'at the same time' is redundant with 'concurrently'.
full	The system should be able to create test environment of weborder system.	It must be possible to create a test environment of the weborder system.	The generated requirement faithfully covers the same core need and preserves all critical details, with only a minor change in modality.
full	Quality Attribute Requirements Upon the USB being plugged in the system shall be able to be deployed and operational in less than 1 minute.	The system must be deployed and fully operational in less than 1 minute upon a USB being plugged in.	The generated requirement faithfully covers the source, preserving all critical details and constraints.
full	The system shall encrypt all sensitive information via https.	All sensitive information must be encrypted via https.	The generated requirement faithfully covers the source, preserving all critical details. The subject 'The system' is implied.
full	The system shall encrypt all customer data in database.	All customer data in the database must be encrypted.	The generated requirement faithfully covers the source, preserving all critical details. The subject 'The system' is implied.
full	Pre condition: A condition that must be satisfied or a state the system must be in before a use case may begin.	A certain condition must be satisfied or the system has to be in a specific state before a use case can begin.	The generated text accurately rephrases the definition of a pre-condition, preserving all critical details and the core meaning.

Verdict	Source requirement	Generated requirement	Judge reason
full	Actors: Customer Weborder System Preconditions: Customer must be able to access the web order system via a web browser with HTTPS.	The customer must be able to access the web order system through a web browser using HTTPS.	The generated text accurately restates the precondition for customer access to the web order system, preserving all critical details.
full	The system shall be able to add product to shopping cart in less than 2ms.	Adding a product to the shopping cart must take less than 2ms.	The generated requirement faithfully rephrases the source, preserving all critical details including the action, object, and performance constraint.

Additional partial-match examples

Source requirement	Generated requirement	Judge reason
Operating Environment Web Store shall operate with the following internet browsers: Microsoft Internet Explorer version 6 and 7, Netscape Communicator Version 4 and 5.	The Web Store must work with Netscape Communicator Version 4.	The generated requirement only specifies one browser version (Netscape 4) out of the four specified in the source (IE 6, 7, Netscape 4, 5).
Web Store shall operate on an Intel based system with Slackware Linux 2.6 and Apache Web Server.	The Web Store must operate on Slackware Linux 2.6.	The generated requirement correctly identifies Slackware Linux 2.6 but omits the Intel based system and Apache Web Server constraints from the source.
Web Store shall operate on an Intel based system with Slackware Linux 2.6 and Apache Web Server.	The Web Store must operate with an Apache Web Server.	The generated requirement correctly identifies Apache Web Server but omits the Intel based system and Slackware Linux 2.6 constraints from the source.
The system shall use SQL based database to store inventory information.	The system must use a SQL based database.	The generated requirement correctly states the use of a SQL based database but omits the critical detail that it is specifically for storing inventory information.
Design and Implementation Constraints Must use a SQL based database.	All inventory information must be stored in the SQL based database.	The generated requirement adds a specific detail about "inventory information" being stored, which is a material addition not present in the general constraint of the source requirement.
The interface must remain consistent among various web browsers and be intuitive to the customer.	The interface has to be intuitive.	The generated requirement omits the critical detail that the interface must also "remain consistent among various web browsers," only covering the intuitiveness aspect.
SI-1.3: Plug-ins interface Communications Interfaces CI-1: The WebOrder system shall send an e-mail confirmation to the customer that the items they ordered will be delivered to the shipping address along with tracking number.	The order confirmation email must include the tracking number.	The generated requirement omits the critical detail that the email confirmation must also state that "items they ordered will be delivered to the shipping address."
The system shall be able to email customer and vendor in less than 1 second.	The system must email both the customer and the vendor after a purchase.	Loses the time constraint and adds a new condition 'after a purchase'.
The system shall be able to validate credit card in less than 2 seconds.	The system must be able to run test credit card transactions.	Changes the purpose from validating credit cards to running test transactions and loses the time constraint.

Source requirement	Generated requirement	Judge reason
The system shall auto detect IP DOS attacks and block IP automatically.	The system must automatically block the IP address from a detected IP DOS attack.	The generated requirement covers blocking IP addresses from detected IP DOS attacks but omits the requirement to automatically detect the attacks.

Document: 1999_dii

Part A - Document header

Document ID	1999_dii
Generation model	gemini / gemini-2.5-pro
Standard validator	gemini / gemini-2.5-flash
Dialogue method	N/A
Question algorithm	N/A
Source requirement count	22
Generated requirement count	22
Matched count	13
Clarification rounds	requested 2 / used 0
Clarification chunks	0
Semantic P / R / F1	0.5909 / 0.5909 / 0.5909
LLM strict P / R / F1	0.1364 / 0.1364 / 0.1364
LLM weighted P / R / F1	0.3409 / 0.3409 / 0.3409
Evidence / Propositions / Gap	32 / 22 / 2
Grounded after rewrite	22

Part B - Generated dialogue transcript

[SYSTEM] To start, what is the overall purpose and scope of this system?

[USER] This project is based on the 1999 - dii system, and the goal is to capture the main requirements and constraints for document 1999_dii.

[SYSTEM] Okay, that helps frame things. Thinking about the actual use of the system, can you describe the key tasks or capabilities different users, like data consumers versus database managers, will need to perform?

[USER] Certainly. We have several different classes of users who will need to use the XS services. There are the 'data consumers', but also our 'data and database managers', and even the 'network information infrastructure resource managers'. A key capability for our managers, particularly when the system is in a 'fixed (static) mode of operation' at a 'base or data processing megacenter', is that our 'on-site personnel' must be able to tune the data management services. They need the ability to 'adjust for varying workloads and sizes of associated databases' to keep things running efficiently.

[SYSTEM] That makes sense. When your on-site personnel are tuning the system for different workloads, how do you envision them confirming that their adjustments are working correctly and haven't introduced any issues with the data?

[USER] That's a critical step. To make sure the adjustments are right and haven't broken anything, our team needs a few ways to verify the system's behavior. We need to be able to qualify the changes through a few different methods. This should include running a test, providing a demonstration, performing analysis on the data, and allowing for a direct inspection of the system.

[SYSTEM] That's a comprehensive approach. When your team is deciding whether to use a test, a demonstration, analysis, or inspection, are there specific situations or types of system adjustments that would call for one method over the others?

[USER] I'm not sure we'd want to choose between them. To make sure we're being thorough, we expect the qualification methods applied to the software to include all of those approaches: test, demonstration, analysis, and inspection. We really need that comprehensive coverage to feel confident.

[SYSTEM] That makes sense for ensuring thorough verification. When your on-site personnel are managing the connections to various databases like the DBIF or DAC, what happens if a database server's name or location changes? How is that information updated within the system?
[USER] That's a great question. If a server name or location changes, our people need the ability to redefine or reset the names of connect descriptors to those database server instances. The key for us is that this should all be transparent to the operator. The system needs to internally interface with the existing data elements of the DBIF, DAC, and our COTS RDBMS products without the user having to worry about the details. And just as a general rule for all the XS components for the DII COE, we have to make sure we adhere to the compliance level requirements described in the I&RTS.
[SYSTEM] That's helpful context on the compliance and transparency aspects. When it comes to the on-site personnel who need to perform these more technical tasks, like redefining connect descriptors, how do you envision they'll be trained to do that, especially since the XS module is embedded within a larger system?
[USER] That's a good question. Since the XS module is just one part of the larger system our people will be using, we think the training requirements really need to be determined by the developers of that main system. They'll have the full picture of how everything fits together and can create a training plan that makes sense.
[SYSTEM] That makes sense to have the main system developers handle the training. Shifting gears slightly to other system management tasks, could you describe how you expect the system to behave when it needs to be taken down for planned maintenance or an upgrade?
[USER] For us, maintenance isn't just one thing. It covers a whole range of activities, including planned shutdowns, upgrades, and modifications. It also means being able to bring parts of the system online or offline, or manage it in specific states like 'open not-mounted'. And speaking of getting the system ready for people, it's important that we have control over session timing. The session time-out values need to be user definable, and we must be able to reset them before a user session is initialized.
[SYSTEM] You've mentioned the need for DII COE compliance a couple of times. When you think about the environments where this system will be deployed, what specific operating systems or platforms does it need to be able to run on to meet those standards?
[USER] That's a really important point. The main thing for us is that the Registry software needs to be portable. It's required to execute on any of our COE-compliant platforms, so we need that flexibility.
[SYSTEM] That makes sense about needing portability across platforms. As the system is used over time, what happens when someone reviews the specifications in the Registry but can't find an existing one to reuse for their specific task?
[USER] That's a good question. If someone has looked through the tags in the Registry and still can't find an existing specification, or 'spec' as we sometimes call them, that they can reuse for their task, then they must create a new one. We can't have people getting stuck, so the process has to be that if you can't find one to reuse, you create a new one.

Part C - Extracted requirements by category

Functional

FR-001 The system shall provide XS services to several different classes of users.

FR-002 The user classes shall include 'data consumers', 'data and database managers', and 'network information infrastructure resource managers'.

FR-003 'On-site personnel' must be able to tune the data management services.

FR-004 The ability to tune data management services must be available when the system is in a 'fixed (static) mode of operation'.

FR-005 The ability to tune data management services must be available at a 'base or data processing megacenter'.

FR-006 Tuning data management services must include the ability to 'adjust for varying workloads and sizes of associated databases'.

FR-007 The qualification methods applied to the software must include all of the approaches: test, demonstration, analysis, and inspection.

FR-008 Personnel must have the ability to redefine or reset the names of connect descriptors to database server instances if a server name or location changes.

FR-009 The system must support bringing parts of the system online or offline.

FR-010 The system must support being managed in specific states like 'open not-mounted'.

FR-011 The session time-out values must be user definable.

FR-012 Session time-out values must be able to be reset before a user session is initialized.

FR-013 The system must provide verification methods to confirm that adjustments are correct and have not introduced any issues.

FR-014 Maintenance activities shall cover planned shutdowns, upgrades, and modifications.

Non-Functional

NFR-001 The ability to redefine or reset the names of connect descriptors to database server instances must be transparent to the operator.

NFR-002 As a general rule, all XS components for the DII COE must adhere to the compliance level requirements described in the I&RTS.

NFR-003 The Registry software is required to execute on any COE-compliant platforms.

Business Rules

BR-001 If a user cannot find an existing specification ('spec') in the Registry to reuse for their task, then they must create a new one.

Interfaces

IF-001 The system must internally interface with the existing data elements of the DBIF.

IF-002 The system must internally interface with the existing data elements of the DAC.

IF-003 The system must internally interface with the existing data elements of COTS RDBMS products.

Constraints

C-001 Training requirements for the XS module are to be determined by the developers of the main system.

Part D - Matched vs missed table

Source requirement text	Matched?	Closest generated requirement text
DOCUMENT OVERVIEW This document outlines the software capabilities required for the XS components for the DII COE.	✓	As a general rule, all XS components for the DII COE must adhere to the compliance level requirements described in the I&RTS.
COE sponsors shall adhere to compliance level requirements described in the I&RTS.	✗	— not recovered
In the fixed (static) mode of operation (base or data processing megacenter), the data management services shall have the capability of being tuned by on-site personnel to adjust for varying workloads and sizes of associated databases.	✓	The ability to tune data management services must be available at a 'base or data processing megacenter'.
The XS shall have the ability to redefine or reset names of connect descriptors to database server instances.	✓	Personnel must have the ability to redefine or reset the names of connect descriptors to database server instances if a server name or location changes.
At a minimum, the session time-out values shall be user definable and be able to be reset prior to initialization of a user session.	✓	Session time-out values must be able to be reset before a user session is initialized.

Source requirement text	Matched?	Closest generated requirement text
End User Mode: Portion of XS services shall be used by various classes of users: data consumers, data and database managers, network information infrastructure resource managers.	✓	The system shall provide XS services to several different classes of users.
Some of these uses of the data management services will entail unique requirements that shall be fulfilled within the capability of XS services.	✗	— not recovered
In addition, maintenance also shall pertain to shutdown, open not-mounted and online/off-line implementations, modifications, upgrade, or other related actions.	✓	Maintenance activities shall cover planned shutdowns, upgrades, and modifications.
The data management services shall support managing various types of data, database architectures and platforms that includes hardware and software at the specified sites.	✓	Tuning data management services must include the ability to 'adjust for varying workloads and sizes of associated databases'.
In support of training activities, the data management services shall provide for the same processing as would be encountered in a production environment.	✓	'On-site personnel' must be able to tune the data management services.
To contribute metadata to a registry, the producer must be able to receive XML registry forms, submit metadata and the related information resource artifacts, and notify.	✗	— not recovered
Extract semantically-valid views from an XML document (e.g., XSL) Disassemble hierarchical view to relational representation Render REGISTRY SERVICES The Registry shall provide standard APIs as specified in Appendix B.	✗	— not recovered
DATA ACCESS SERVICES INTERNAL INTERFACE REQUIREMENTS The XS function shall provide standard APIs as specified in Appendix C.	✗	— not recovered
The XS shall internally interface (transparent to the operator) with the existing data elements of the DBIF, DAC, and COTS RDBMS products.	✓	The system must internally interface with the existing data elements of the DBIF.
REGISTRY SERVICES ENVIRONMENT REQUIREMENTS The Registry software shall be portable and required to execute on COE-compliant platforms.	✓	The Registry software is required to execute on any COE-compliant platforms.
Personnel requirements shall be determined by the developers of the system in which the XS module is embedded TRAINING-RELATED REQUIREMENTS Not applicable.	✗	— not recovered
Training requirements shall be determined by the developers of the system in which the XS module is embedded.	✓	Training requirements for the XS module are to be determined by the developers of the main system.
OTHER REQUIREMENTS None PACKAGING REQUIREMENTS The XS software shall be delivered in accordance with DII COE guidelines.	✗	— not recovered
The Qualification Methods applied to the software shall include test, demonstration, analysis, and inspection (T, D, A, I).	✓	The qualification methods applied to the software must include all of the approaches: test, demonstration, analysis, and inspection.

Source requirement text	Matched?	Closest generated requirement text
Facts are either links (relationships) between entities or associations between an entity and attributes Constraints are conditions about the data that must always be true.	✗	— not recovered
If, after reviewing the tags in the Registry, you cannot reuse an existing specification (a.k.a.	✓	If a user cannot find an existing specification ('spec') in the Registry to reuse for their task, then they must create a new one.
But if they were identified as attributes of the element ctry, then the additional baggage (other attributes) must be carried, or submitted as separate elements.	✗	— not recovered

Part E - Gemini judge requirement matching

Strict P/R/F1: 0.1364 / 0.1364 / 0.1364. Weighted P/R/F1: 0.3409 / 0.3409 / 0.3409.

Judge scoring: strict counts only full matches. Weighted counts full=1 and partial=0.5.

Verdict	Source requirement	Generated requirement	Judge reason
full	The Qualification Methods applied to the software shall include test, demonstration, analysis, and inspection (T, D, A, I).	The qualification methods applied to the software must include all of the approaches: test, demonstration, analysis, and inspection.	The generated requirement accurately restates the source, preserving all critical details regarding the qualification methods and their mandatory inclusion.
full	Training requirements shall be determined by the developers of the system in which the XS module is embedded.	Training requirements for the XS module are to be determined by the developers of the main system.	The generated requirement accurately reflects that the main system developers determine the XS module's training requirements, preserving all critical details.
full	If, after reviewing the tags in the Registry, you cannot reuse an existing specification (a.k.a.	If a user cannot find an existing specification ('spec') in the Registry to reuse for their task, then they must create a new one.	The generated requirement faithfully completes the implied rule from the incomplete source text, preserving the core condition and providing a logical consequence.
partial	The XS shall have the ability to redefine or reset names of connect descriptors to database server instances.	The ability to redefine or reset the names of connect descriptors to database server instances must be transparent to the operator.	The generated requirement adds a new material condition ('transparent to the operator') and loses the specific actor 'XS' from the source.
partial	REGISTRY SERVICES ENVIRONMENT REQUIREMENTS The Registry software shall be portable and required to execute on COE-compliant platforms.	The Registry software is required to execute on any COE-compliant platforms.	The generated requirement loses the critical detail that the Registry software 'shall be portable'.
partial	The XS shall internally interface (transparent to the operator) with the existing data elements of the DBIF, DAC, and COTS RDBMS products.	The system must internally interface with the existing data elements of COTS RDBMS products.	The generated requirement covers internal interfacing with COTS RDBMS products but loses the specific mention of 'DBIF', 'DAC', and 'transparent to the operator', and generalizes 'XS' to 'system'.
partial	At a minimum, the session time-out values shall be user definable and be able to be reset prior to initialization of a user session.	The session time-out values must be user definable.	The generated requirement loses the critical detail that session time-out values must also 'be able to be reset prior to initialization of a user session'.

Verdict	Source requirement	Generated requirement	Judge reason
partial	End User Mode: Portion of XS services shall be used by various classes of users: data consumers, data and database managers, network information infrastructure resource managers.	The user classes shall include 'data consumers', 'data and database managers', and 'network information infrastructure resource managers'.	The generated requirement loses the context that these user classes will use a 'Portion of XS services' in 'End User Mode'. It only lists the classes.
partial	COE sponsors shall adhere to compliance level requirements described in the I&RTS.	As a general rule, all XS components for the DII COE must adhere to the compliance level requirements described in the I&RTS.	The generated requirement changes the subject from 'COE sponsors' to 'XS components for the DII COE', which is a material alteration of the actor responsible for adherence.
partial	In the fixed (static) mode of operation (base or data processing megacenter), the data management services shall have the capability of being tuned by on-site personnel to adjust for varying workloads and sizes of associated databases.	Tuning data management services must include the ability to 'adjust for varying workloads and sizes of associated databases'.	The generated requirement loses critical details regarding the mode of operation (fixed/static), the location (base or data processing megacenter), and the actor (on-site personnel) who performs the tuning.
partial	DATA ACCESS SERVICES INTERNAL INTERFACE REQUIREMENTS The XS function shall provide standard APIs as specified in Appendix C.	The system shall provide XS services to several different classes of users.	The generated requirement covers 'XS services' but loses the critical details of 'standard APIs' and 'Appendix C' and adds 'to several different classes of users'.
partial	In addition, maintenance also shall pertain to shutdown, open not-mounted and online/off-line implementations, modifications, upgrade, or other related actions.	The system must support being managed in specific states like 'open not-mounted'.	The generated requirement preserves one specific state ('open not-mounted') but loses other critical details about maintenance actions and states mentioned in the source (shutdown, online/off-line, modifications, upgrade).

Additional partial-match examples

Source requirement	Generated requirement	Judge reason
In the fixed (static) mode of operation (base or data processing megacenter), the data management services shall have the capability of being tuned by on-site personnel to adjust for varying workloads and sizes of associated databases.	The ability to tune data management services must be available when the system is in a 'fixed (static) mode of operation'.	The generated requirement loses critical details about the purpose of tuning (adjusting for workloads/database sizes), the actor (on-site personnel), and the specific location (base or data processing megacenter).
In the fixed (static) mode of operation (base or data processing megacenter), the data management services shall have the capability of being tuned by on-site personnel to adjust for varying workloads and sizes of associated databases.	The ability to tune data management services must be available at a 'base or data processing megacenter'.	The generated requirement loses critical details about the purpose of tuning (adjusting for workloads/database sizes), the actor (on-site personnel), and the mode of operation (fixed/static).
In the fixed (static) mode of operation (base or data processing megacenter), the data management services shall have the capability of being tuned by on-site personnel to adjust for varying workloads and sizes of associated databases.	'On-site personnel' must be able to tune the data management services.	The generated requirement loses the context of 'fixed (static) mode of operation' and the purpose of tuning 'to adjust for varying workloads and sizes of associated databases'.
The XS shall have the ability to redefine or reset names of connect descriptors to database server instances.	Personnel must have the ability to redefine or reset the names of connect descriptors to database server instances if a server name or location changes.	The generated requirement changes the actor from 'The XS' to 'Personnel' and adds a new material condition ('if a server name or location changes').

Source requirement	Generated requirement	Judge reason
At a minimum, the session time-out values shall be user definable and be able to be reset prior to initialization of a user session.	Session time-out values must be able to be reset before a user session is initialized.	The generated requirement loses the critical detail that session time-out values must also be 'user definable'.
End User Mode: Portion of XS services shall be used by various classes of users: data consumers, data and database managers, network information infrastructure resource managers.	The system shall provide XS services to several different classes of users.	The generated requirement generalizes 'Portion of XS services' to 'XS services' and omits the specific examples of user classes (data consumers, data and database managers, network information infrastructure resource managers).
In addition, maintenance also shall pertain to shutdown, open not-mounted and online/off-line implementations, modifications, upgrade, or other related actions.	Maintenance activities shall cover planned shutdowns, upgrades, and modifications.	The generated requirement covers some maintenance activities (shutdowns, upgrades, modifications) but omits others (open not-mounted, online/off-line implementations, other related actions) and adds 'planned' to shutdowns.
The XS shall internally interface (transparent to the operator) with the existing data elements of the DBIF, DAC, and COTS RDBMS products.	The system must internally interface with the existing data elements of the DBIF.	The generated requirement loses the 'transparent to the operator' condition and only specifies one of the three required interfaces (DBIF, DAC, COTS RDBMS).
The XS shall internally interface (transparent to the operator) with the existing data elements of the DBIF, DAC, and COTS RDBMS products.	The system must internally interface with the existing data elements of the DAC.	The generated requirement loses the 'transparent to the operator' condition and only specifies one of the three required interfaces (DBIF, DAC, COTS RDBMS).
OTHER REQUIREMENTS None PACKAGING REQUIREMENTS The XS software shall be delivered in accordance with DII COE guidelines.	As a general rule, all XS components for the DII COE must adhere to the compliance level requirements described in the I&RTS.	The generated requirement shifts from 'delivery in accordance with DII COE guidelines' to 'adherence to compliance level requirements described in the I&RTS' for DII COE components, losing the 'delivery' aspect and adding specific compliance details.

Document: 2005_microcare

Part A - Document header

Document ID	2005_microcare
Generation model	gemini / gemini-2.5-pro
Standard validator	gemini / gemini-2.5-flash
Dialogue method	N/A
Question algorithm	N/A
Source requirement count	25
Generated requirement count	36
Matched count	21
Clarification rounds	requested 2 / used 2
Clarification chunks	2
Semantic P / R / F1	0.5833 / 0.8400 / 0.6885
LLM strict P / R / F1	0.1667 / 0.2400 / 0.1967
LLM weighted P / R / F1	0.3611 / 0.5200 / 0.4262
Evidence / Propositions / Gap	38 / 36 / 1
Grounded after rewrite	36

Part B - Generated dialogue transcript

[SYSTEM] To start, what is the overall purpose and scope of this system?

[USER] This project is based on the 2005 - microcare system, and the goal is to capture the main requirements and constraints for document 2005_microcare.

[SYSTEM] Great, that helps set the scope. Thinking about the functions from that 2005 system, what specific details would you need to capture for each salesperson in their master record?

[USER] Yes, that's a key part of it. We need to have a 'Salesman master' record, and it's important that it captures all the necessary information for each person.

[SYSTEM] Since those Salesman master records are so important, what kind of controls do we need around who can view or edit them, and should we keep a history of any changes?

[USER] Yes, absolutely. We need to make sure that only an authorized user can alter or amend the claim if required. And it's also important that during the claim entry process, the system captures the necessary information about what's happening.

[SYSTEM] Okay, that's clear on capturing the right information. During that claim entry process, when someone is creating an invoice, how should the system verify that the total amount is calculated correctly based on the number of vouchers and their price?

[USER] That's a good question. The calculation itself is simple: the invoice amount should just be the number of vouchers multiplied by the wholesale price. For these transactions, the mode of payment is always Cash. To make sure we're tracking the right vouchers, each voucher slip should have a bar code with two identifications, one for the client and one for the partner. Speaking of getting the data right, there are a few other things. When a user is entering VSP details, we want to make sure the data is consistent. The values for Health Sub – District, Locality, Level Of Facility, Type of facility, and Registration Body should come from their own master list, so the user has to select the details based on the VSP. Also, when creating a new staff member, if a staff with an existing name is entered, the system should generate a message to warn the user, but it's important that the system will allow the user to save that new staff if it's required.

[SYSTEM] That's very helpful for ensuring data consistency. Thinking more about the vouchers themselves, besides the price, are there other key details like an expiration date that the system needs to validate when a claim is processed?
[USER] Yes, absolutely. The validity date is a key detail, and it's important that it can only be amended or altered at the system level by an authorized user. The voucher will also have an MRP, or maximum retail price. Generally, all the rules for the voucher, like the health care services included and the quality standards for providers, are defined in the PDS. The physical voucher is also designed with three tear off portioned slips, and one has a sub-section tear-off for the partner. If a claim doesn't meet these rules, our user needs to be able to mark the Claim status as 'Rejected' and send it back to the VSP.
[SYSTEM] That's clear on how to handle rejected claims. For a successful claim, you mentioned printing a receipt. What specific information from the claim submission should be automatically included on that receipt?
[USER] Yes, the receipt should be based on the information from the claim submission. A critical part for us is tracking the vouchers. The receipt needs to automatically list the details for each voucher used, pulling the `Batch No` and `voucher No` from our available voucher stock. It would be great if this was shown in a grid format to make it clear. We also need to include the `Patient Status`, which should be selected from our list in the `Patient Status Master`. If the status happens to be 'Referral', then it's very important that the receipt also captures who the patient was `Referred To` and the `Reason for Referral`, with that reason being populated from our `Referral master`.
[SYSTEM] You've mentioned a few times that only an 'authorized user' can make certain changes, like amending voucher validity. Could you tell me more about the different roles you see in the system? For instance, what would a system administrator be able to do that a standard claims processor could not?
[USER] That's a good question. A system admin would have much higher-level permissions. For instance, only a system admin should be able to create the User Groups we use to manage access. We'll need the flexibility to create any number of individual users under any of those user groups. A regular user, on the other hand, would be doing more of the daily tasks. For example, during every distribution transaction, they would need to select the name of the salesman from our Team Master list.
[SYSTEM] We've discussed entering VSP details and processing their claims. When you need to look up a VSP's master file, what key information, such as the agreed-upon payment terms, would you expect to see on that summary screen?
[USER] When we pull up a VSP's master file, it's critical that we can see all the key information for that partner. Specifically, the VSP Master Information screen needs to show the payment terms we've agreed upon between MSIU and the VSP. We need a place to enter those terms and then have them clearly displayed on that summary.
[SYSTEM] That's clear on the VSP master file. Thinking about how the VSPs will actually submit claims, I recall a mention of using thumb impressions for verification. Will they need to use a specific type of fingerprint scanner or device at their locations to interact with the system?
[USER] That's an important point. We're concerned about fraud, so if a claim from the same VSP has a thumb impression mismatch more than two times, we need the system to automatically show a warning message. At the same time, it should also in-activate that VSP to prevent any further submissions.

Part C - Extracted requirements by category

Functional

FR-001	When creating a new staff member, if a staff with an existing name is entered, the system must generate a warning message.
FR-002	The system must allow a user to save a new staff member with an existing name if required.
FR-003	A user must be able to mark a claim's status as 'Rejected'.
FR-004	The system must allow a rejected claim to be sent back to the VSP.
FR-005	The receipt must be based on information from the claim submission.
FR-006	The receipt must automatically list the details for each voucher used.
FR-007	The system needs to provide the flexibility to create any number of individual users under any of the user groups.
FR-008	During every distribution transaction, the user must select the name of the salesman from the Team Master list.

FR-009 The system needs a place to enter payment terms agreed upon between MSIU and the VSP, which are then displayed on the VSP Master Information screen.

Non-Functional

NFR-001 Only an authorized user can alter or amend a claim.

NFR-002 During the claim entry process, the system must capture the necessary information about what's happening.

NFR-003 The voucher validity date can only be amended or altered at the system level by an authorized user.

NFR-004 The details for each voucher used on a receipt should be shown in a grid format.

NFR-005 Only a system admin can create User Groups.

Data

DATA-001 The system needs to have a 'Salesman master' record that captures all the necessary information for each person.

DATA-002 The 'Salesman master' record must capture all necessary information for each person.

DATA-003 Each voucher slip should have a bar code with two identifications, one for the client and one for the partner.

DATA-004 The value for 'Health Sub – District' must be selected from a master list.

DATA-005 When a user is entering VSP details, the value for 'Locality' should come from its own master list.

DATA-006 When a user is entering VSP details, the value for 'Level Of Facility' should come from its own master list.

DATA-007 When a user is entering VSP details, the value for 'Registration Body' should come from its own master list.

DATA-008 The voucher will have an MRP, or maximum retail price.

DATA-009 The receipt needs to automatically list the details for each voucher used, pulling the `Batch No` and `voucher No` from the available voucher stock.

DATA-010 The receipt must include the `Patient Status`.

DATA-011 The `Patient Status` must be selected from the `Patient Status Master` list.

DATA-012 The `Reason for Referral` must be populated from the `Referral master`.

DATA-013 The value for 'Locality' must be selected from a master list.

DATA-014 The value for 'Level Of Facility' must be selected from a master list.

Business Rules

BR-001 The invoice amount shall be calculated as the number of vouchers multiplied by the wholesale price.

BR-002 For transactions where the invoice amount is the number of vouchers multiplied by the wholesale price, the mode of payment is always Cash.

BR-003 If the `Patient Status` is 'Referral', the receipt must capture who the patient was `Referred To`.

BR-004 If a claim from the same VSP has a thumb impression mismatch more than two times, the system must in-activate that VSP.

Interfaces

IF-001 Each voucher slip must have a bar code.

Constraints

C-001	All rules for the voucher, including health care services and quality standards for providers, are defined in the PDS.
C-002	The physical voucher must be designed with three tear-off portioned slips.
C-003	One of the voucher's tear-off slips must have a sub-section tear-off for the partner.

Part D - Matched vs missed table

Source requirement text	Matched?	Closest generated requirement text
The principles of the voucher, the contents of the health care services, their required quality, the standards to be met by the providers and their infrastructure, as well as the measures for monitoring the quality have been defined in the PDS.	✓	All rules for the voucher, including health care services and quality standards for providers, are defined in the PDS.
Each voucher should have two portions with three tear off voucher slips each for Client and Partner.	✓	The physical voucher must be designed with three tear-off portioned slips.
Each voucher should have the following properties, which will have sub-elements to get the batch numbers, voucher numbers, and the project codes.	✓	The details for each voucher used on a receipt should be shown in a grid format.
This validity date can be amended or altered at the system level ONLY by the authorized user Voucher will also have MRP (maximum retail price) Voucher should have three tear off portioned slips with a sub-section tear-off for the partner.	✓	The voucher validity date can only be amended or altered at the system level by an authorized user.
Each voucher slip should contain the bar code of the Voucher with two identifications one for the client and the other for the partner.	✓	Each voucher slip should have a bar code with two identifications, one for the client and one for the partner.
During every distribution transaction the user should select the name of the sales man listed from Team Master.	✓	During every distribution transaction, the user must select the name of the salesman from the Team Master list.
The Salesman master should capture following information's.	✓	The 'Salesman master' record must capture all necessary information for each person.
Name of the distributor* Name of the Sales Man* Date of distribution* No of vouchers sold* According to the number of vouchers required by the distributor, the system should allocate the relevant vouchers with their numbers and batch numbers based on the stock.	✓	The receipt must automatically list the details for each voucher used.
Invoice amount = No of vouchers x Wholesale price Mode of payment is Cash The mandatory information required during a distribution transaction is mentioned below.	✓	For transactions where the invoice amount is the number of vouchers multiplied by the wholesale price, the mode of payment is always Cash.
While entering the No of vouchers required, the system should automatically pick the Batch No's with voucher No's from the available voucher stock and list the details of each voucher with below information's as a grid format.	✓	The receipt needs to automatically list the details for each voucher used, pulling the `Batch No` and `voucher No` from the available voucher stock.
The values of Health Sub – District, Locality, Level Of Facility, Type of facility, Registration Body would be list from their own master and should select the details based on the VSP.	✓	When a user is entering VSP details, the value for 'Level Of Facility' should come from its own master list.
The VSP Master Information screen should have a provision to enter the payment terms agreed between MSIU and VSP.	✓	The system needs a place to enter payment terms agreed upon between MSIU and the VSP, which are then displayed on the VSP Master Information screen.

Source requirement text	Matched?	Closest generated requirement text
VSP Name Staff Code Staff Name Staff Type Qualification The system will generate message while creating a new staff with existing name, but the system will allow the user to save that new staff if its required.	✓	The system must allow a user to save a new staff member with an existing name if required.
Claim (Treatment Form) Submission Depends on the payment terms (Bi-monthly / Monthly) mentioned in the VSP master the VSP should submit the Claim (Treatment form) to the MISU Field office.	✗	— not recovered
These information is vital and shall be used for compared with the processed claims.	✗	— not recovered
In the transaction part no of forms should match with No of Available Forms The system should print a receipt document based on the information entered during claim submission.	✓	The receipt must be based on information from the claim submission.
The document should have the following information.	✗	— not recovered
Before the claim entry the user should check the form manually to understand whether any mandatory information is missed in the Claim or not.	✗	— not recovered
If yes, then the user should mark that Claim (Treatment Form) status as Rejected and send back to VSP.	✓	The system must allow a rejected claim to be sent back to the VSP.
During the claim entry the system should capture the following information.	✓	During the claim entry process, the system must capture the necessary information about what's happening.
Patient Status - (Select from list of status options populating from Patient Status Master) suppose the status is Referral then the system will have the provision to capture Referred To and Reason for Referral, the reason for Referral should be populated from Referral master.	✓	If the \`Patient Status\` is 'Referral', the receipt must capture who the patient was \`Referred To\`.
The system should have a provision to capture the reason suppose if the claim status is “Rejection or Quarantine” Claim Amount – The calculation of claim amount is explained below.	✓	A user must be able to mark a claim's status as 'Rejected'.
If the claim from same VSP having more than two times of Thump Impression mismatching, then the system should automatically produce a warning message and the same time the system should In-Activate the VSP.	✓	If a claim from the same VSP has a thumb impression mismatch more than two times, the system must in-activate that VSP.
The authorized user can alter / amend the Claim if required.	✓	Only an authorized user can alter or amend a claim.
The system admin only can able to create User Groups The system should have the provision to create any number of individual user under any user group.	✓	Only a system admin can create User Groups.

Part E - Gemini judge requirement matching

Strict P/R/F1: 0.1667 / 0.2400 / 0.1967. Weighted P/R/F1: 0.3611 / 0.5200 / 0.4262.

Judge scoring: strict counts only full matches. Weighted counts full=1 and partial=0.5.

Verdict	Source requirement	Generated requirement	Judge reason
full	Each voucher slip should contain the bar code of the Voucher with two identifications one for the client and the other for the partner.	Each voucher slip should have a bar code with two identifications, one for the client and one for the partner.	The generated requirement faithfully covers the source, preserving all critical details about the barcode and its two identifications.
full	The authorized user can alter / amend the Claim if required.	Only an authorized user can alter or amend a claim.	The generated requirement faithfully captures that only an authorized user can alter or amend a claim, which is the core intent of the source. The 'if required' condition from the source is a contextual detail for when the action happens, not who can perform it.
full	During every distribution transaction the user should select the name of the sales man listed from Team Master.	During every distribution transaction, the user must select the name of the salesman from the Team Master list.	The generated requirement faithfully covers the source, preserving all critical details including the action, actor, context, and data source.
full	This validity date can be amended or altered at the system level ONLY by the authorized user Voucher will also have MRP (maximum retail price) Voucher should have three tear off portioned slips with a sub-section tear-off for the partner.	The voucher validity date can only be amended or altered at the system level by an authorized user.	The generated requirement accurately and completely captures the detail about the voucher validity date being amended or altered only by an authorized user at the system level.
full	The Salesman master should capture following information’s.	The 'Salesman master' record must capture all necessary information for each person.	The generated requirement faithfully covers the source, interpreting 'following information’s' as 'all necessary information' which is a reasonable and faithful interpretation given the source's vagueness.
full	Patient Status - (Select from list of status options populating from Patient Status Master) suppose the status is Referral then the system will have the provision to capture Referred To and Reason for Referral, the reason for Referral should be populated from Referral master.	The `Reason for Referral` must be populated from the `Referral master`.	The generated requirement faithfully captures the detail that 'Reason for Referral' must be populated from 'Referral master'.
partial	The VSP Master Information screen should have a provision to enter the payment terms agreed between MSIU and VSP.	The system needs a place to enter payment terms agreed upon between MSIU and the VSP, which are then displayed on the VSP Master Information screen.	The generated requirement alters the location of the entry provision, implying the entry might not be directly on the VSP Master Information screen, and adds a display aspect.
partial	During the claim entry the system should capture the following information.	During the claim entry process, the system must capture the necessary information about what's happening.	The generated requirement generalizes 'the following information' to 'the necessary information about what's happening', losing specificity.
partial	If the claim from same VSP having more than two times of Thump Impression mismatching, then the system should automatically produce a warning message and the same time the system should In-Activate the VSP.	If a claim from the same VSP has a thumb impression mismatch more than two times, the system must in-activate that VSP.	The generated requirement misses one of the two critical actions specified in the source: producing a warning message.

Verdict	Source requirement	Generated requirement	Judge reason
partial	The system admin only can able to create User Groups The system should have the provision to create any number of individual user under any user group.	Only a system admin can create User Groups.	The generated requirement covers the first part of the source (system admin creating user groups) but completely omits the second part about creating individual users under groups.
partial	VSP Name Staff Code Staff Name Staff Type Qualification The system will generate message while creating a new staff with existing name, but the system will allow the user to save that new staff if its required.	The system must allow a user to save a new staff member with an existing name if required.	The generated requirement only covers the saving aspect and omits the critical detail about generating a message when creating a new staff with an existing name.
partial	Each voucher should have two portions with three tear off voucher slips each for Client and Partner.	One of the voucher's tear-off slips must have a sub-section tear-off for the partner.	The generated requirement significantly alters the described physical structure of the voucher, changing 'two portions with three tear off voucher slips each for Client and Partner' to 'one of the voucher's tear-off slips must have a sub-section tear-off for the partner'.
partial	The principles of the voucher, the contents of the health care services, their required quality, the standards to be met by the providers and their infrastructure, as well as the measures for monitoring the quality have been defined in the PDS.	All rules for the voucher, including health care services and quality standards for providers, are defined in the PDS.	The generated requirement generalizes several specific details from the source, such as 'required quality' of services, 'infrastructure' for providers, and completely omits 'measures for monitoring the quality'.
partial	While entering the No of vouchers required, the system should automatically pick the Batch No's with voucher No's from the available voucher stock and list the details of each voucher with below information's as a grid format.	The receipt needs to automatically list the details for each voucher used, pulling the `Batch No` and `voucher No` from the available voucher stock.	The generated requirement loses the condition 'While entering the No of vouchers required' and the display format 'as a grid format' mentioned in the source.
partial	Invoice amount = No of vouchers x Wholesale price Mode of payment is Cash The mandatory information required during a distribution transaction is mentioned below.	For transactions where the invoice amount is the number of vouchers multiplied by the wholesale price, the mode of payment is always Cash.	The generated requirement makes the cash payment conditional on the invoice calculation, which is not explicitly stated in the source. It also completely omits the part about mandatory information.

Additional partial-match examples

Source requirement	Generated requirement	Judge reason
Each voucher should have two portions with three tear off voucher slips each for Client and Partner.	The physical voucher must be designed with three tear-off portioned slips.	The generated requirement loses critical details about the voucher's structure, specifically 'two portions' and the explicit allocation 'each for Client and Partner'.
Each voucher should have the following properties, which will have sub-elements to get the batch numbers, voucher numbers, and the project codes.	The voucher will have an MRP, or maximum retail price.	The source specifies properties like batch numbers, voucher numbers, and project codes. The generated requirement introduces MRP as a property, which is a new material detail, and misses the specific properties mentioned in the source.

Source requirement	Generated requirement	Judge reason
This validity date can be amended or altered at the system level ONLY by the authorized user Voucher will also have MRP (maximum retail price) Voucher should have three tear off portioned slips with a sub-section tear-off for the partner.	One of the voucher's tear-off slips must have a sub-section tear-off for the partner.	The source specifies 'three tear off portioned slips with a sub-section tear-off for the partner'. The generated loses the critical detail about the total number of slips ('three') and implies only 'one' slip has the sub-section.
Each voucher slip should contain the bar code of the Voucher with two identifications one for the client and the other for the partner.	Each voucher slip must have a bar code.	The generated requirement loses the critical detail about the two identifications (client and partner) associated with the barcode.
Name of the distributor* Name of the Sales Man* Date of distribution* No of vouchers sold* According to the number of vouchers required by the distributor, the system should allocate the relevant vouchers with their numbers and batch numbers based on the stock.	The receipt needs to automatically list the details for each voucher used, pulling the `Batch No` and `voucher No` from the available voucher stock.	The generated requirement correctly identifies 'Batch No' and 'voucher No' related to 'voucher stock' as mentioned in the source. However, it introduces the concept of a 'receipt' and 'vouchers used', which are not explicitly part of the source's core need for *allocating* vouchers.
Invoice amount = No of vouchers x Wholesale price Mode of payment is Cash The mandatory information required during a distribution transaction is mentioned below.	The invoice amount shall be calculated as the number of vouchers multiplied by the wholesale price.	The generated requirement only covers the invoice amount calculation and omits the details about the mode of payment and mandatory information from the source.
While entering the No of vouchers required, the system should automatically pick the Batch No’s with voucher No’s from the available voucher stock and list the details of each voucher with below information’s as a grid format.	The receipt must automatically list the details for each voucher used.	The generated requirement is a significant generalization, losing critical details such as the trigger condition ('While entering the No of vouchers required'), the specific data to be picked ('Batch No’s with voucher No’s from the available voucher stock'), and the display format ('as a grid format').
While entering the No of vouchers required, the system should automatically pick the Batch No’s with voucher No’s from the available voucher stock and list the details of each voucher with below information’s as a grid format.	The details for each voucher used on a receipt should be shown in a grid format.	The generated requirement only captures the display format ('grid format') and loses the context of when this listing occurs ('While entering the No of vouchers required') and how the data is obtained ('system should automatically pick the Batch No’s with voucher No’s from the available voucher stock').
The values of Health Sub – District, Locality, Level Of Facility, Type of facility, Registration Body would be list from their own master and should select the details based on the VSP.	When a user is entering VSP details, the value for 'Registration Body' should come from its own master list.	The generated requirement only covers one of the listed fields and adds a user action context, while omitting the "based on VSP" selection condition.
The values of Health Sub – District, Locality, Level Of Facility, Type of facility, Registration Body would be list from their own master and should select the details based on the VSP.	When a user is entering VSP details, the value for 'Locality' should come from its own master list.	The generated requirement only covers one of the listed fields and adds a user action context, while omitting the "based on VSP" selection condition.